# Translation: From RNA to Protein

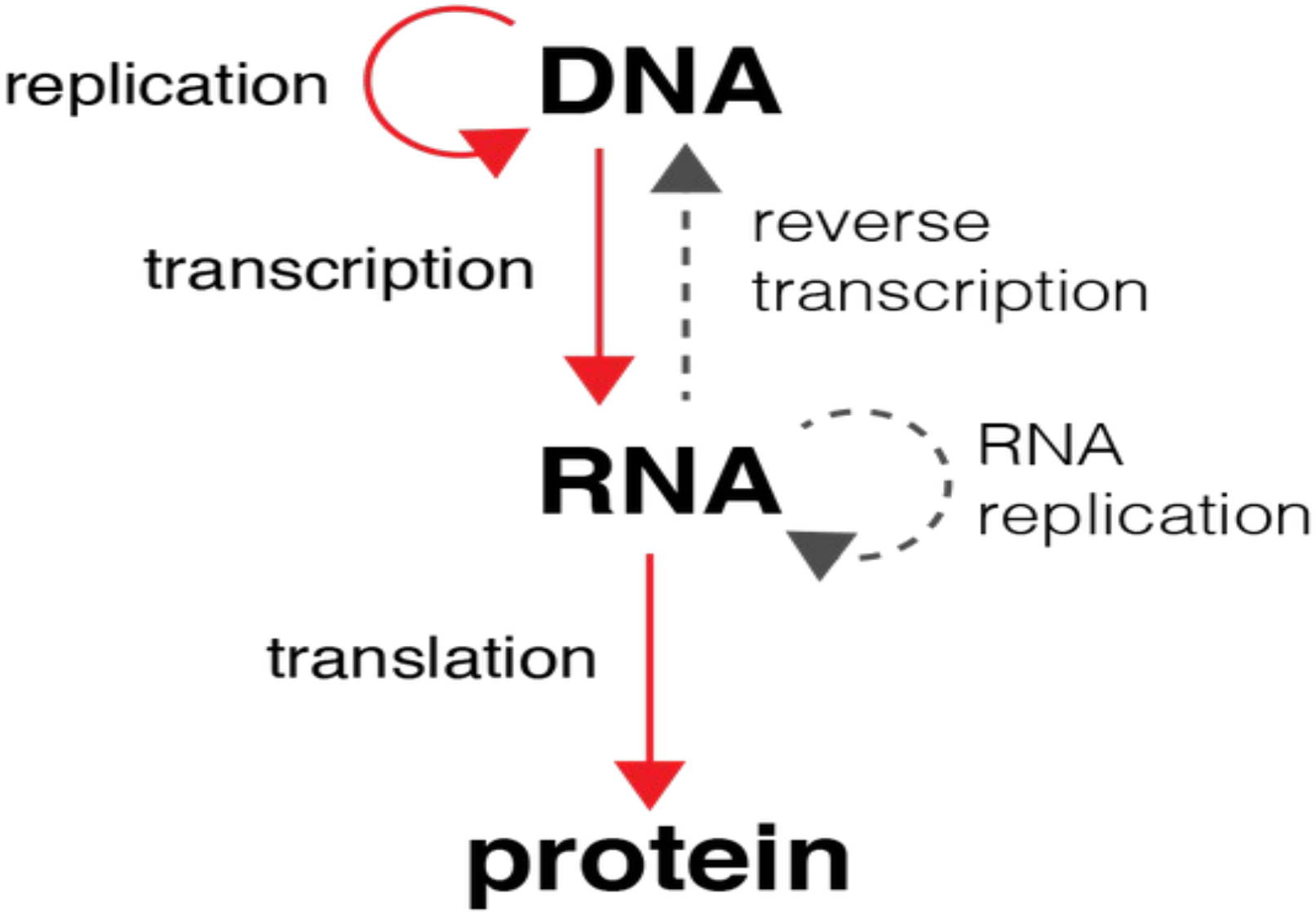
# Lecture Objectives

**At the end of this session students will be able to understand;**

**1.The role of protein translation in the central dogma of life.**

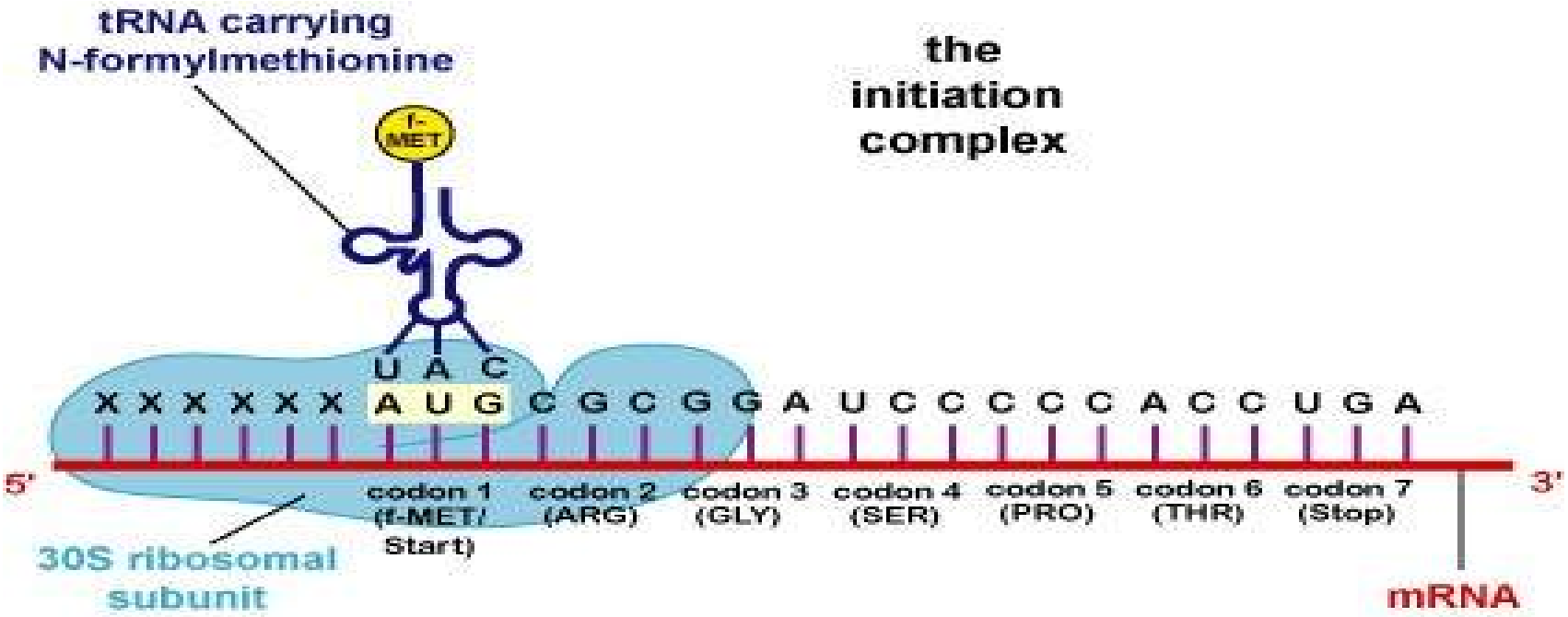
**2.What is protein translation and the prominent steps involved in this mechanism**

**3.Importance of Protein translation and post translational modifications involved in it.**

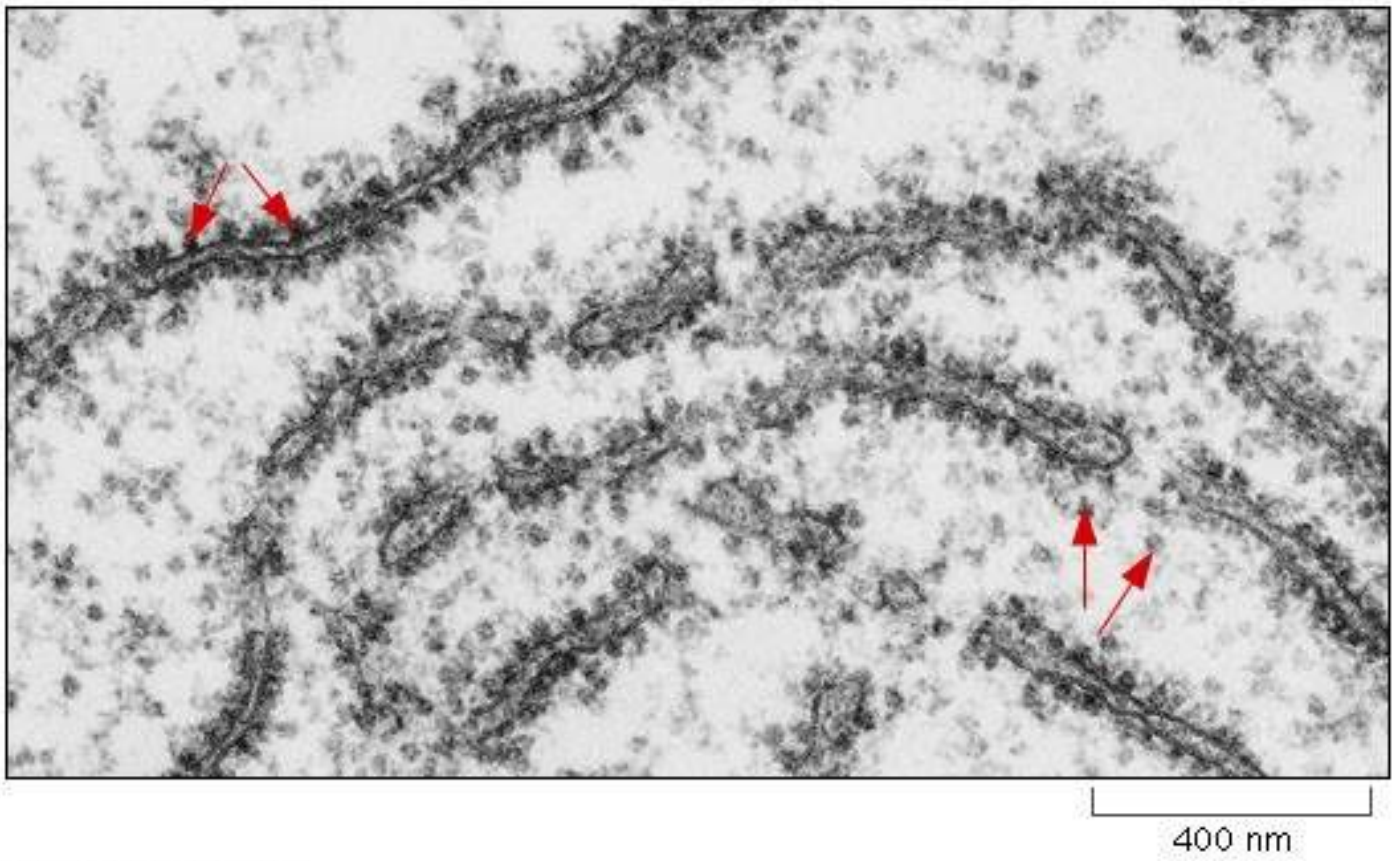

replication
DNA
transcription
reverse
transcription
RNA
RNA
replication
translation
protein

# Translation

**Translation is the process by which protein is made from an mRNA template.**

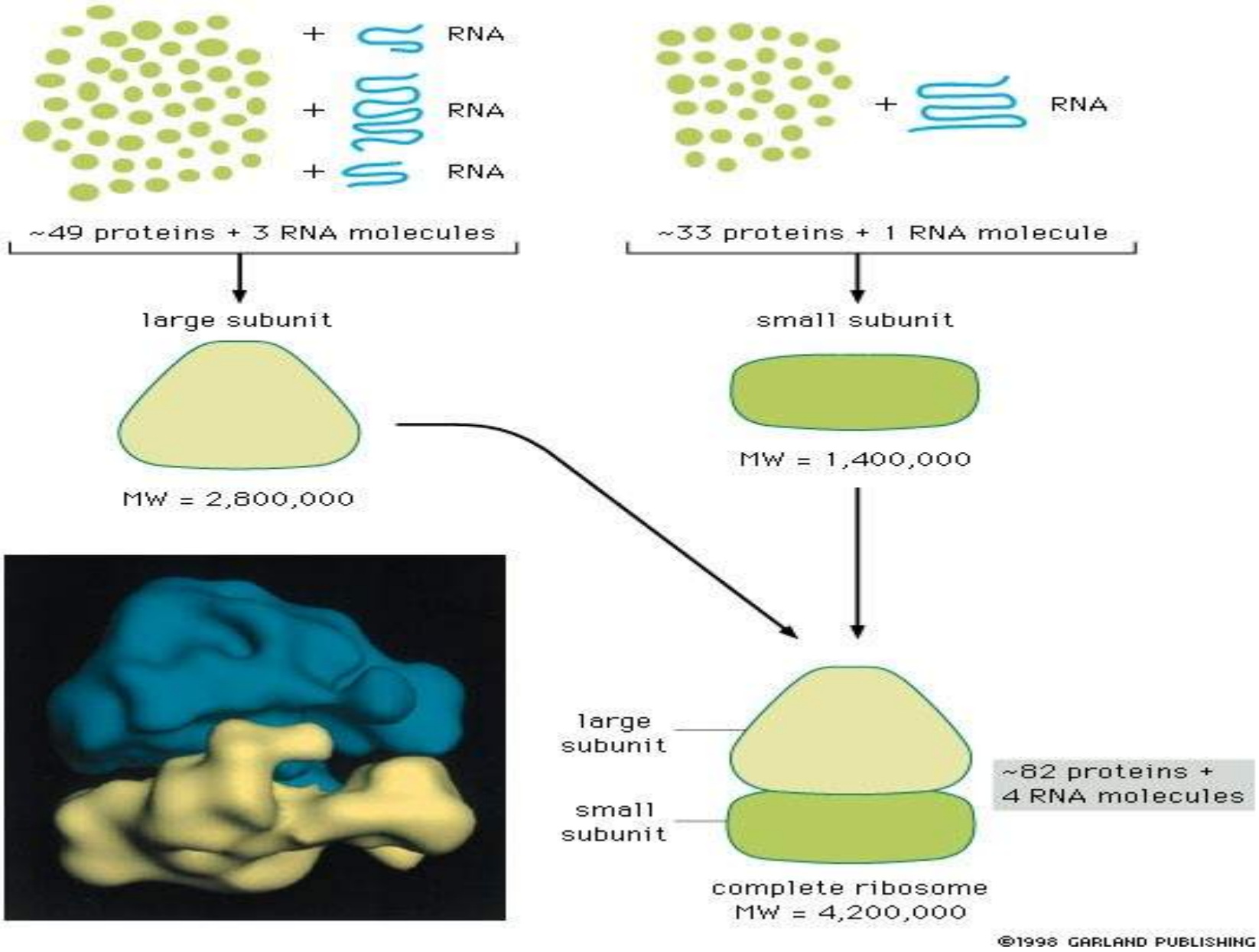


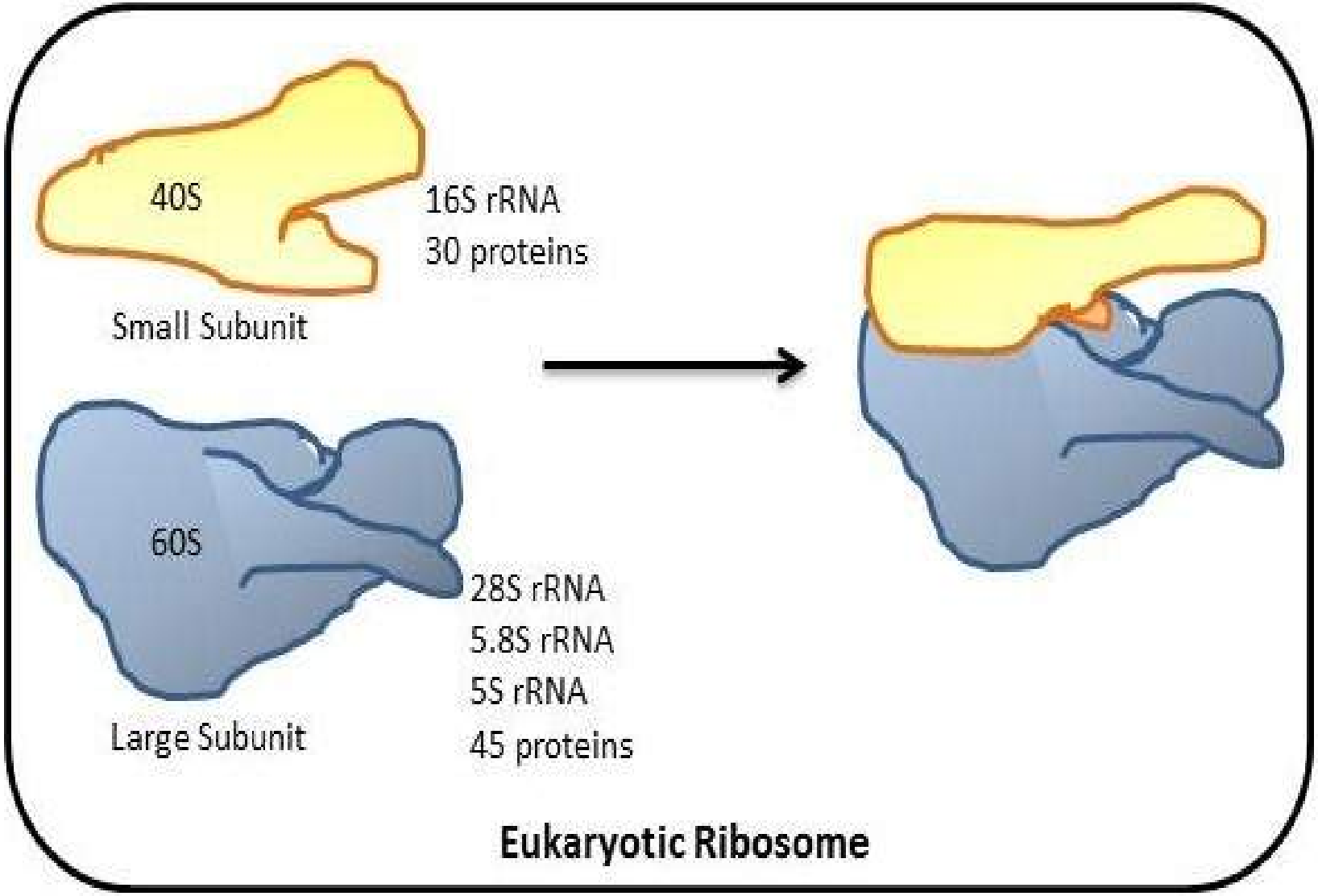
# Ribosomes in the cytoplasm of a eukaryotic cell

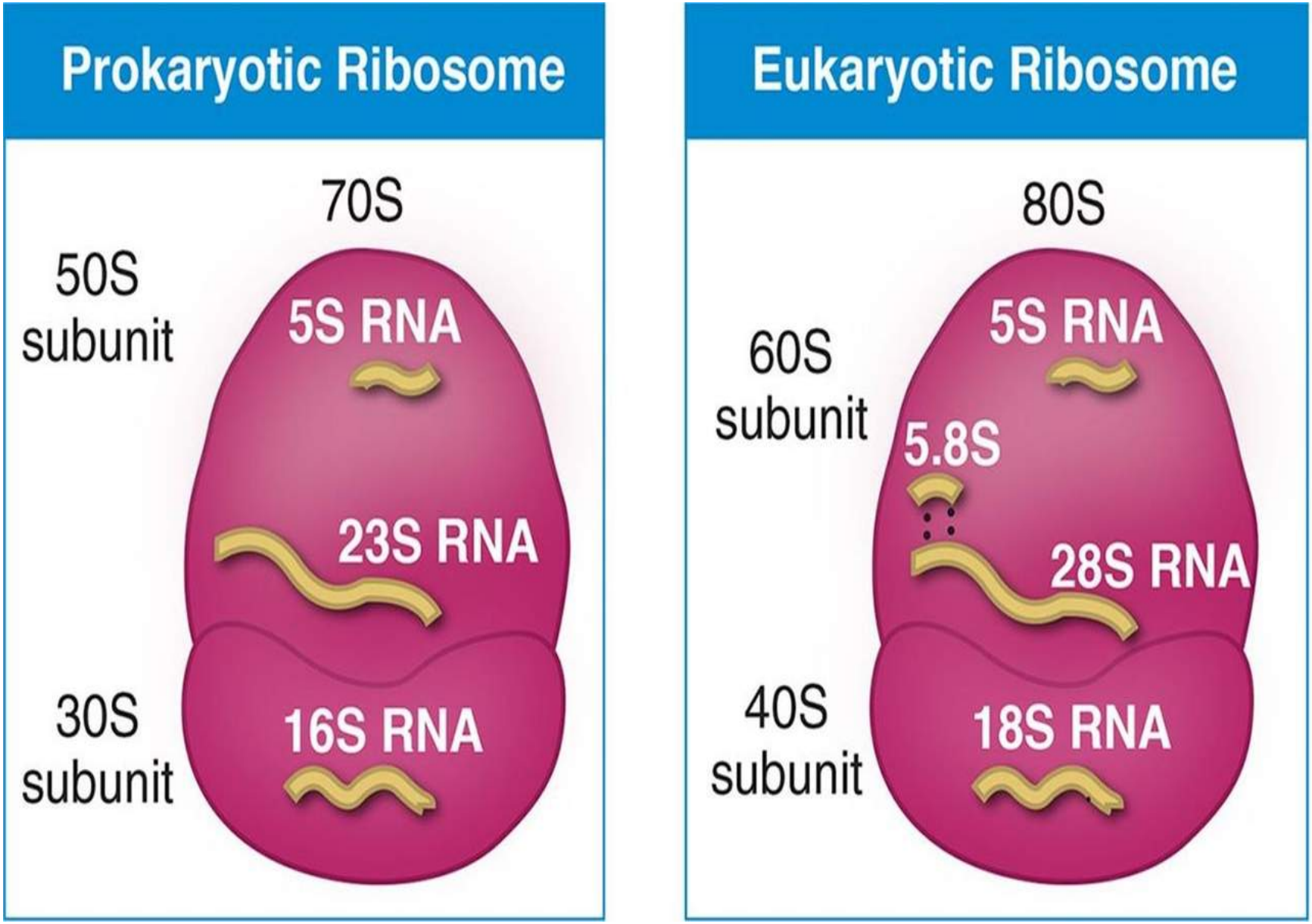



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# COMPOSITION OF EUKARYOTIC RIBOSOMES




40S
16S rRNA
30 proteins
Small Subunit
60S
28S rRNA
5.8S rRNA
5S rRNA
45 proteins
Large Subunit
Eukaryotic Ribosome


Prokaryotic Ribosome
70S
50S subunit
5S RNA
23S RNA
30S subunit
16S RNA
Eukaryotic Ribosome
80S
60S subunit
5S RNA
5.8S
28S RNA
40S subunit
18S RNA

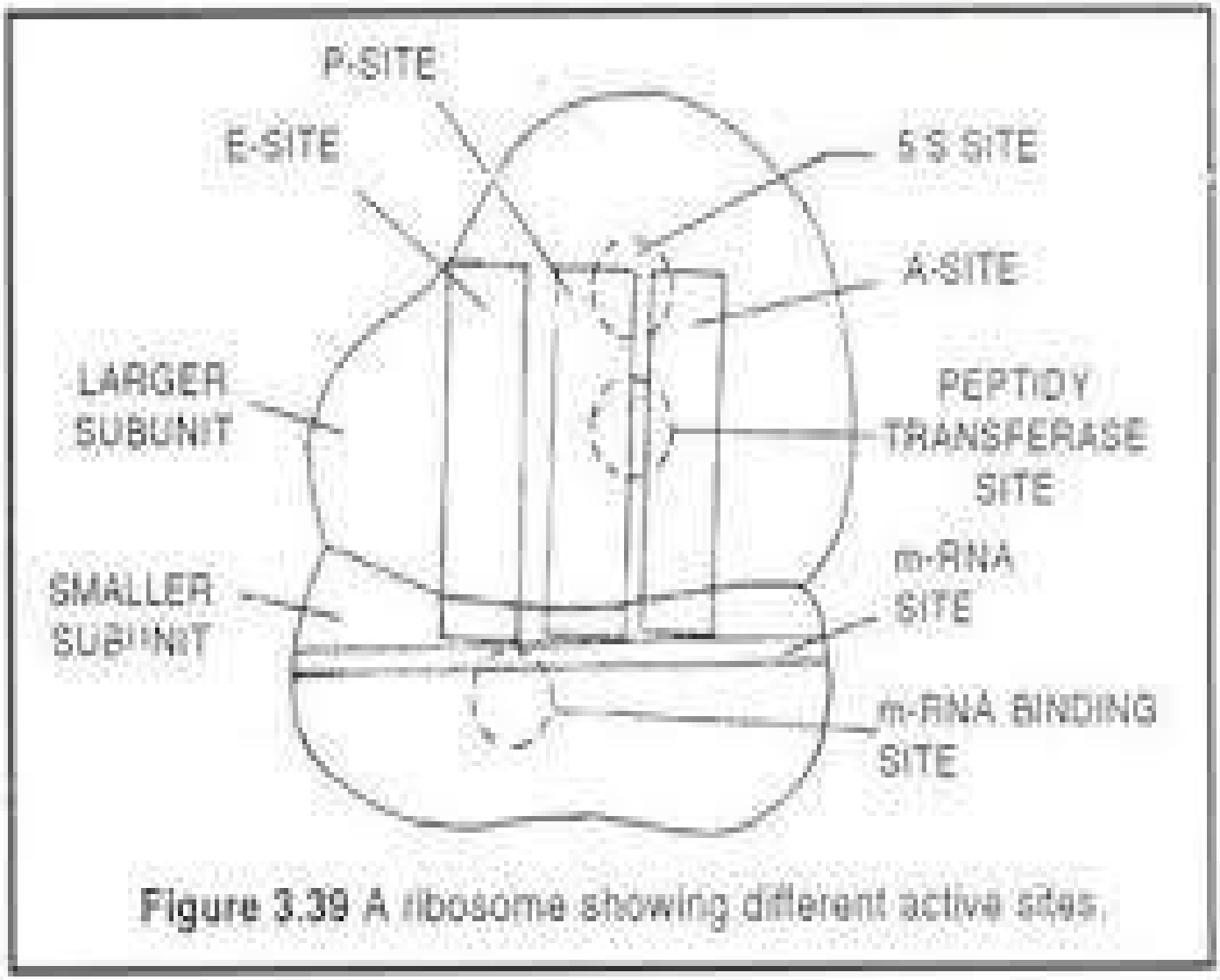
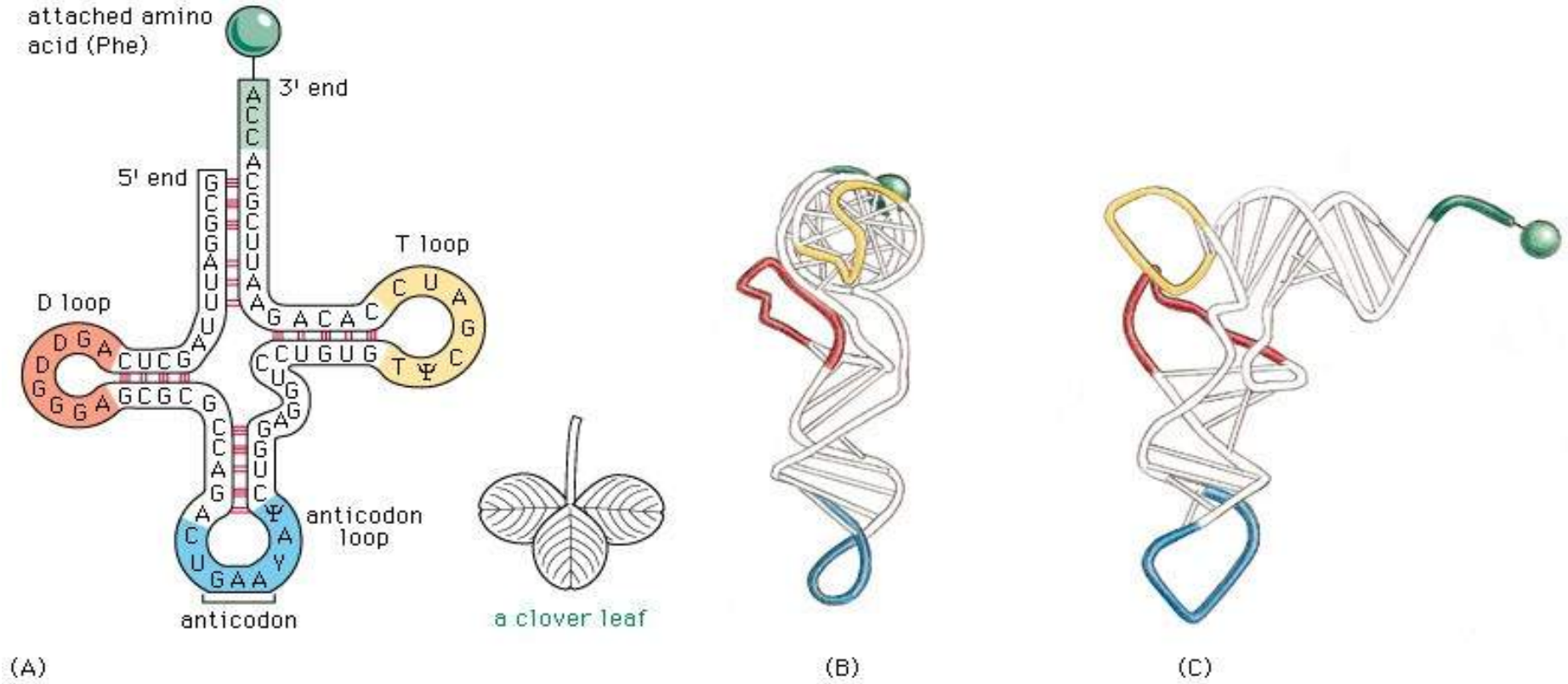



**Figure 3.39** A ribosome showing different active sites.

# tRNA molecules: matching amino acids to codons in mRNA



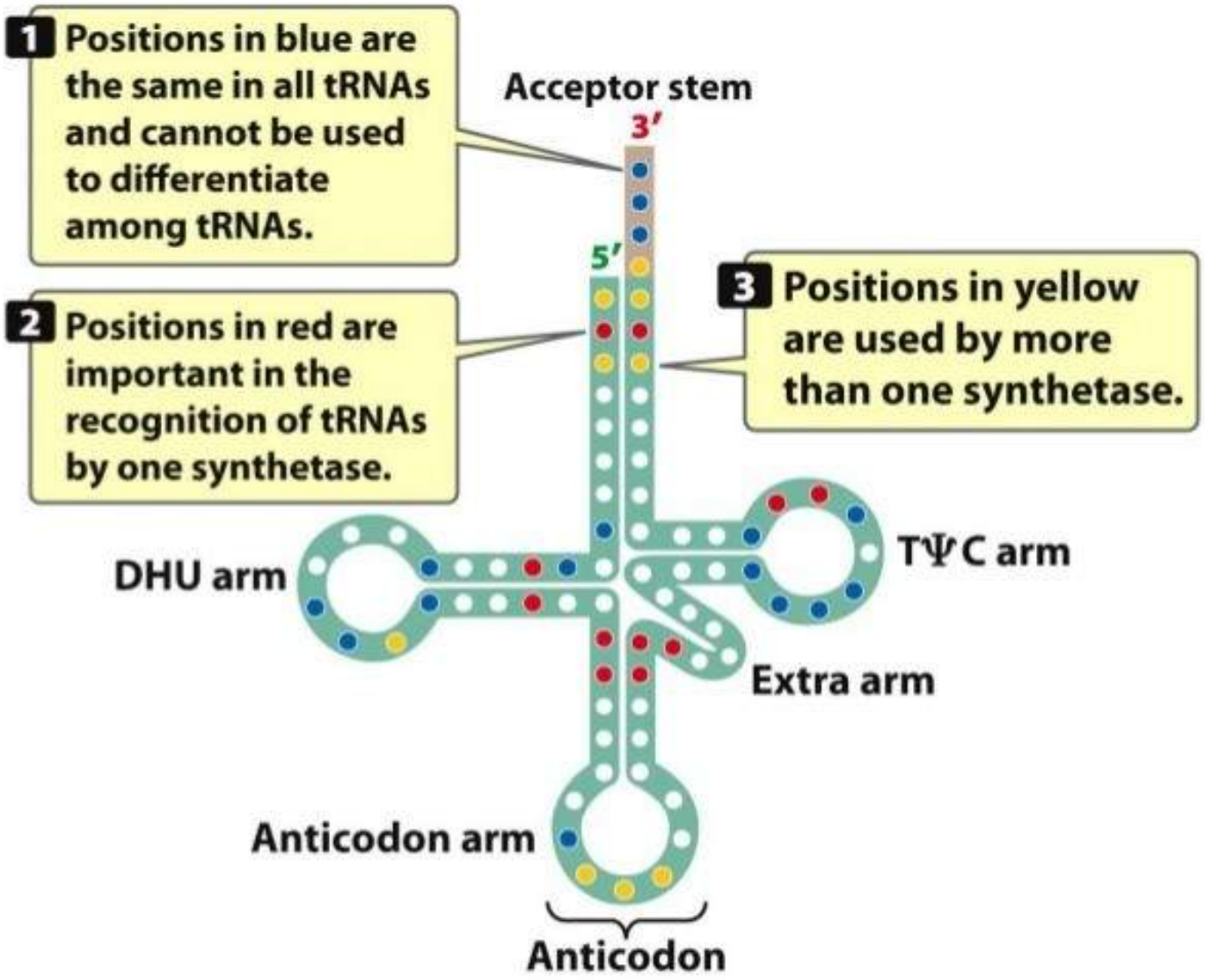
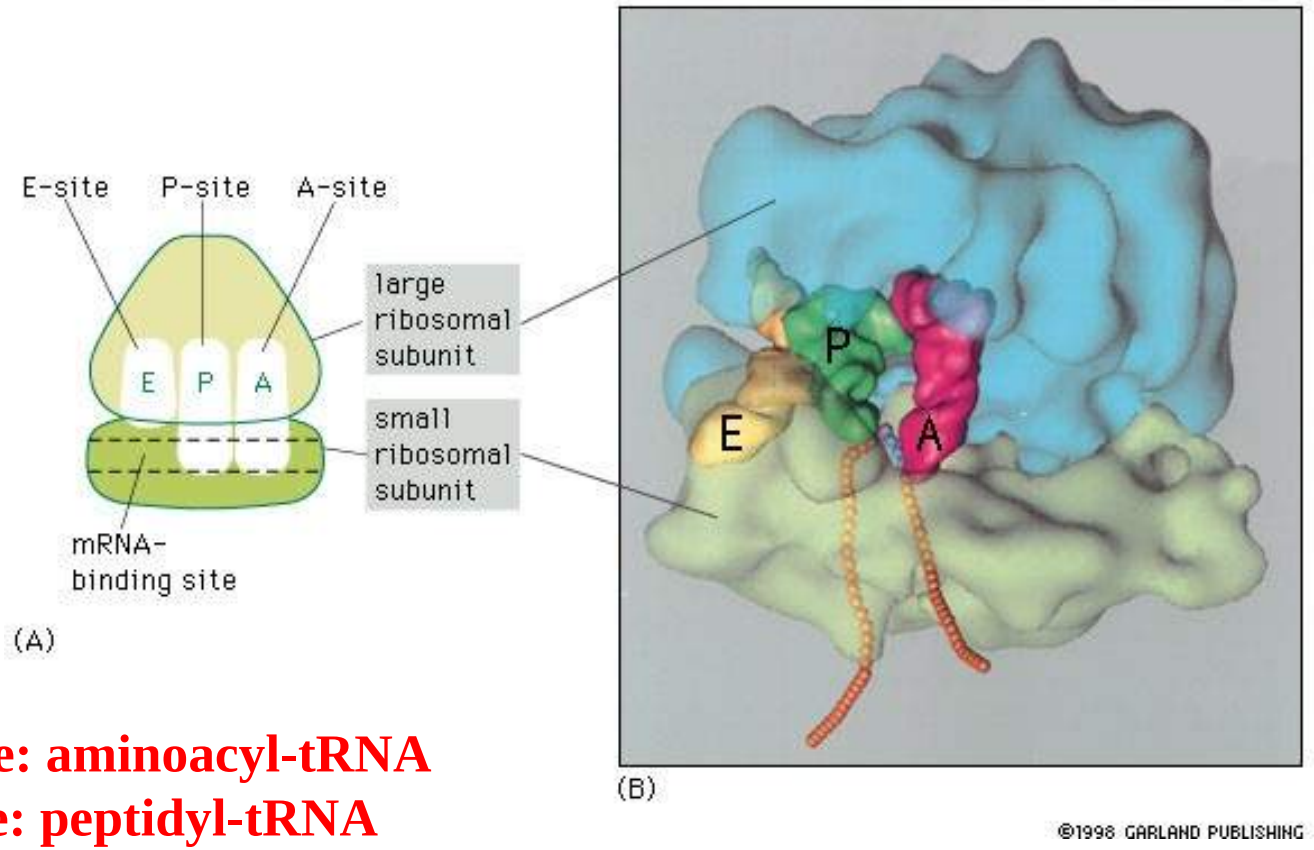



**Figure 15.14**
*Genetics: A Conceptual Approach*, Fourth Edition
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# RNA-binding sites in the ribosome

**Each ribosome has:**

- **a binding site for mRNA**
- **three binding sites for tRNA**




- **A-site: aminoacyl-tRNA**
- **P-site: peptidyl-tRNA**
- **E-site: exit**

# The Genetic Code

- **Codon:** **The three nucleotide triplet base sequence serve as a code word for amino acid synthesis.**
- **Nucleotide bases includes: bases U, G, A, C.**
- **These four bases produce 64 different combinations ($4^3$). (61 for only 20 a.a)**
- **The Nucleotide sequence of codon on mRNA is written from 5' end to 3' end.**

| 1st letter | Second Letter: U | | Second Letter: C | | Second Letter: A | | Second Letter: G | | 3rd letter |
|---|---|---|---|---|---|---|---|---|---|
| U | UUU | Phe | UCU | Ser | UAU | Tyr | UGU | Cys | U |
| | UUC | | UCC | | UAC | | UGC | | C |
| | UUA | Leu | UCA | | UAA | Stop | UGA | Stop | A |
| | UUG | | UCG | | UAG | Stop | UGG | Trp | G |
| C | CUU | Leu | CCU | Pro | CAU | His | CGU | Arg | U |
| | CUC | | CCC | | CAC | | CGC | | C |
| | CUA | | CCA | | CAA | Gln | CGA | | A |
| | CUG | | CCG | | CAG | | CGG | | G |
| A | AUU | Ile | ACU | Thr | AAU | Asn | AGU | Ser | U |
| | AUC | | ACC | | AAC | | AGC | | C |
| | AUA | | ACA | | AAA | Lys | AGA | Arg | A |
| | AUG | Met | ACG | | AAG | | AGG | | G |
| G | GUU | Val | GCU | Ala | GAU | Asp | GGU | Gly | U |
| | GUC | | GCC | | GAC | | GGC | | C |
| | GUA | | GCA | | GAA | Glu | GGA | | A |
| | GUG | | GCG | | GAG | | GGG | | G |

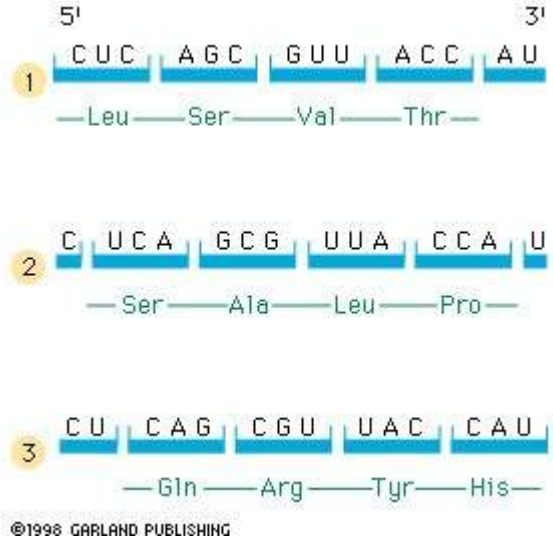
# From RNA to protein: translation

## The genetic code

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| GCA<br>GCC<br>GCG<br>GCU | AGA<br>AGG<br>CGA<br>CGC<br>CGG<br>CGU | GAC<br>GAU | AAC<br>AAU | UGC<br>UGU | GAA<br>GAG | CAA<br>CAG | GGA<br>GGC<br>GGG<br>GGU | CAC<br>CAU | AUA<br>AUC<br>AUU | UUA<br>UUG<br>CUA<br>CUC<br>CUG<br>CUU | AAA<br>AAG | AUG | UUC<br>UUU | CCA<br>CCC<br>CCG<br>CCU | AGC<br>AGU<br>UCA<br>UCC<br>UCG<br>UCU | ACA<br>ACC<br>ACG<br>ACU | UGG | UAC<br>UAU | GUA<br>GUC<br>GUG<br>GUU | UAA<br>UAG<br>UGA |
| Ala | Arg | Asp | Asn | Cys | Glu | Gln | Gly | His | Ile | Leu | Lys | Met | Phe | Pro | Ser | Thr | Trp | Tyr | Val | stop |
| A | R | D | N | C | E | Q | G | H | I | L | K | M | F | P | S | T | W | Y | V | |

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## Three possible “reading frames”



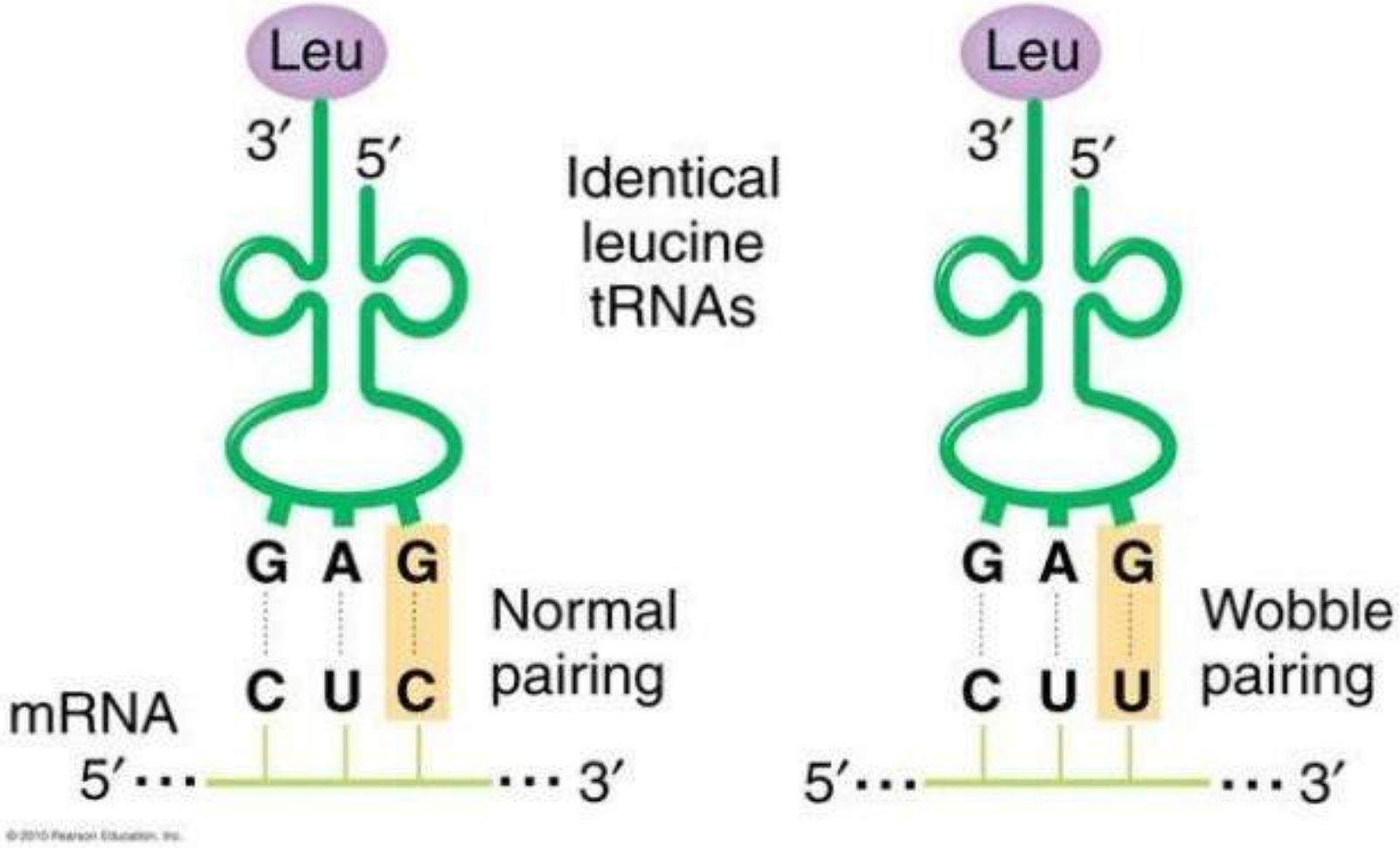

**THE ABC FOR THE DNA**

**Insertion (X)**
**or**
**Deletion (B)**

**THE AXB CFO RTH EDN A**

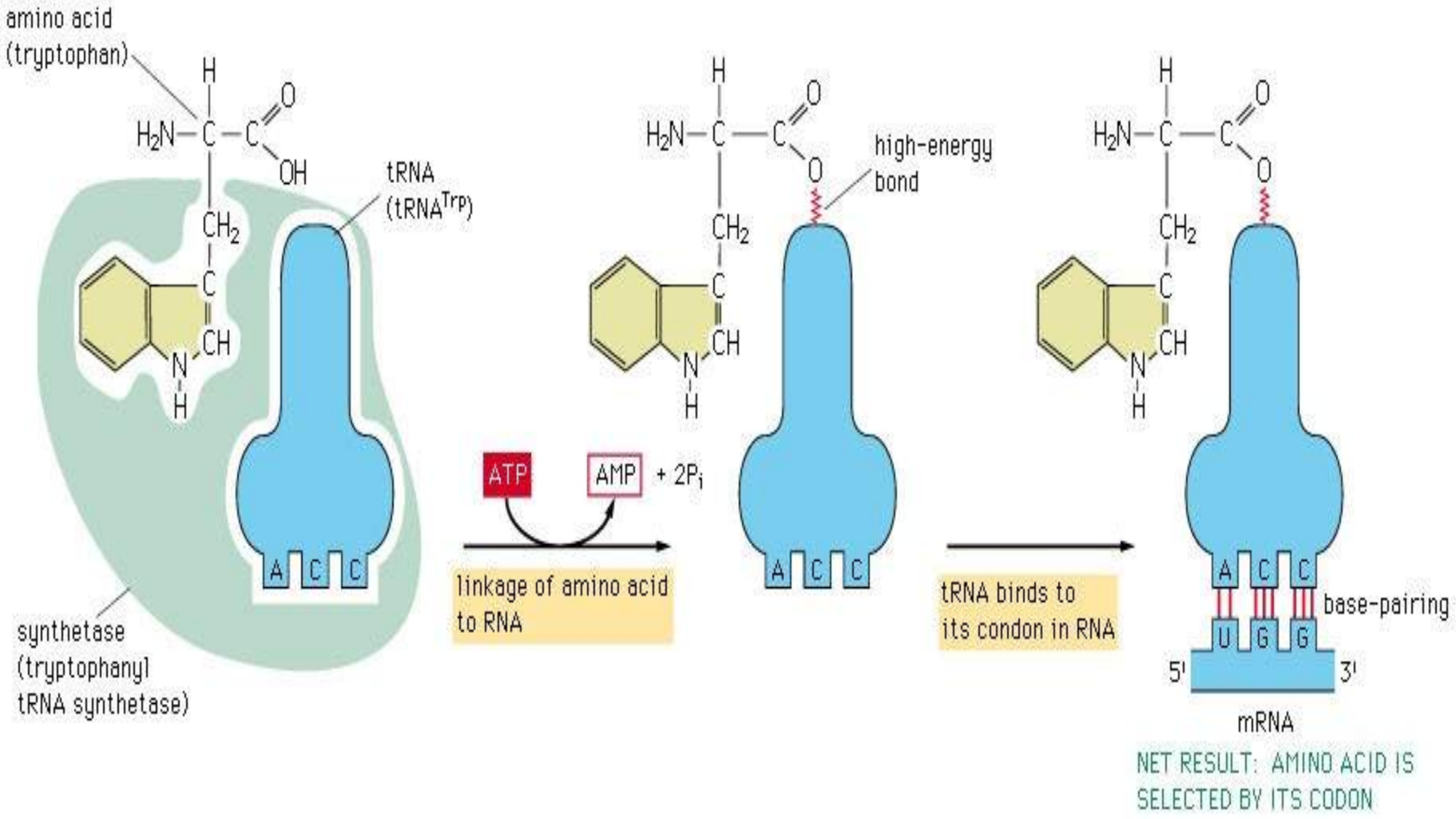
**THE ACF ORT HED NA**

# Wobble Hypothesis




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# Translation of the genetic code: two adaptors that act one after another

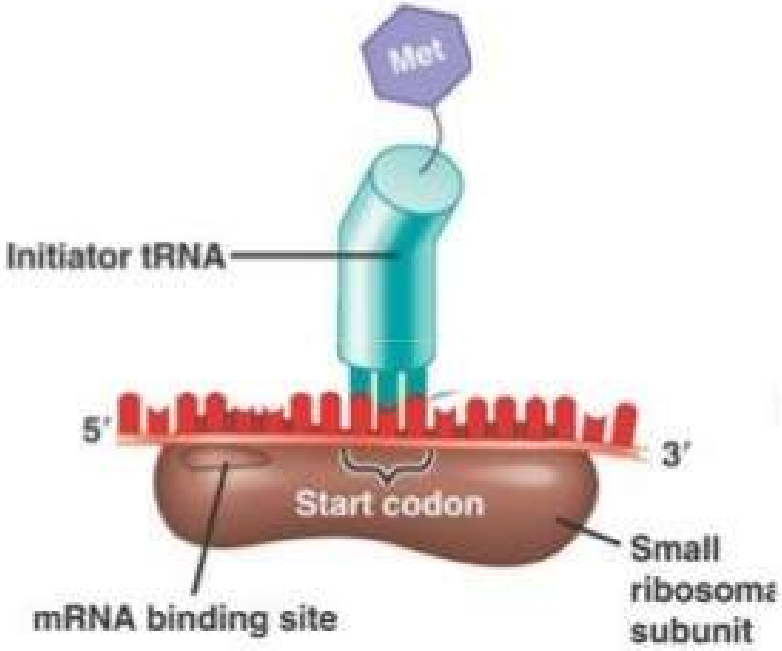


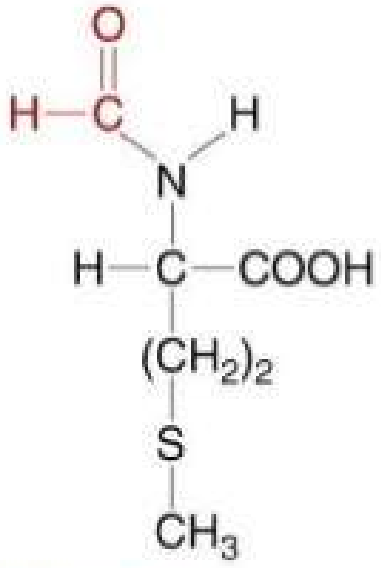

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# Initiation step 3: Initator tRNA binds

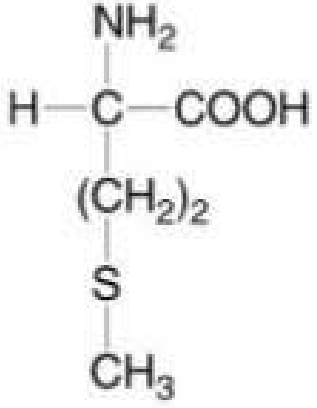
Starting amino acid:

- Prokaryotes: formyl-methionine (fmet)
- Eukaryotes: regular methionine (met)




*N*-Formylmethionine




Methionine

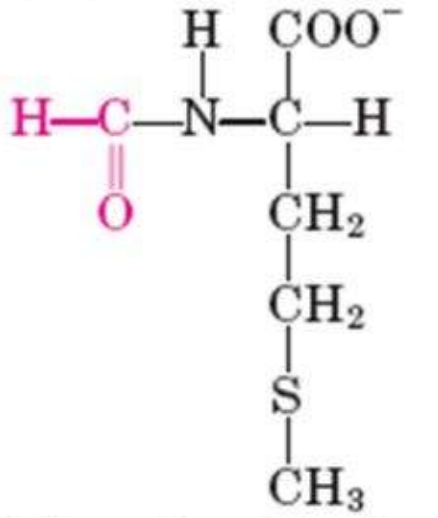
# Prokaryotic Met-tRNA$^{met}$

- **Prokaryotic Met-tRNA$^{met}$ can be formylated to fMet-tRNA$_i^{met}$.**

**Met-tRNA$^{met}$ + N$^{10}$-formyl tetrahydrofolate**

↓ formyl transferase

**fMet-tRNA$_i^{met}$ + tetrahydrofolate**



*N*-Formylmethionine

# 24.6 A Special Initiator tRNA Starts the Polypeptide Chain

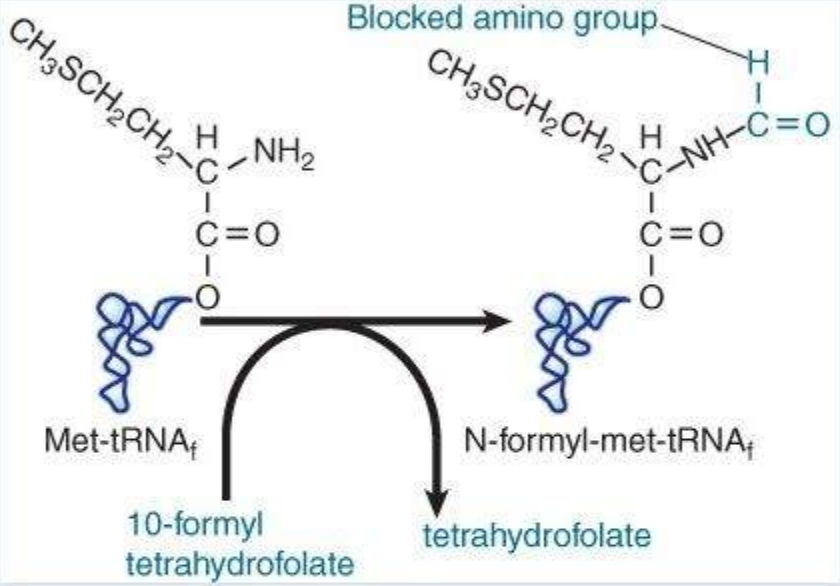



Figure 24.14: The initiator N-formyl-methionyl-tRNA is generated by formylation of methionyl-tRNA, using formyl-tetrahydrofolate as cofactor.

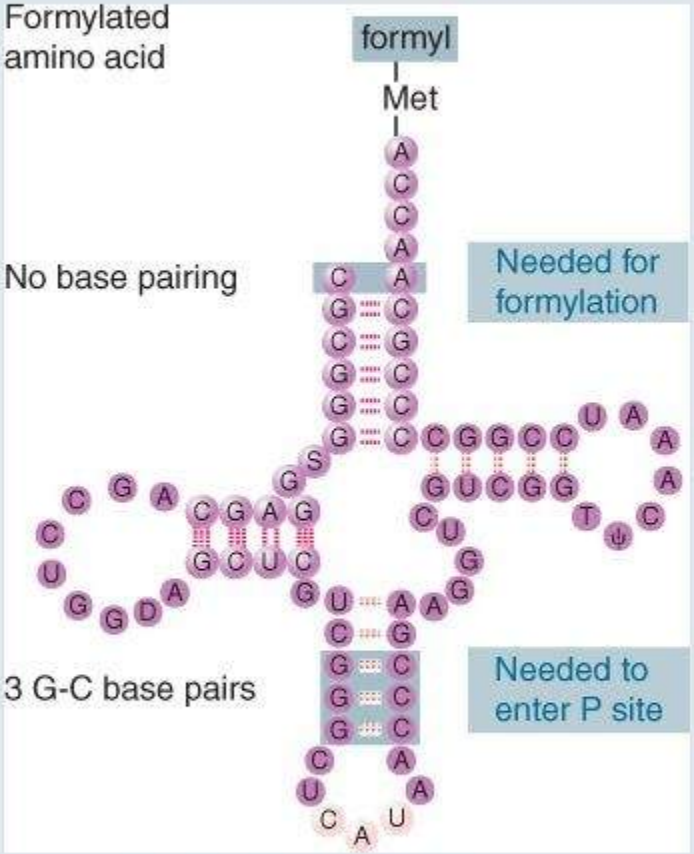
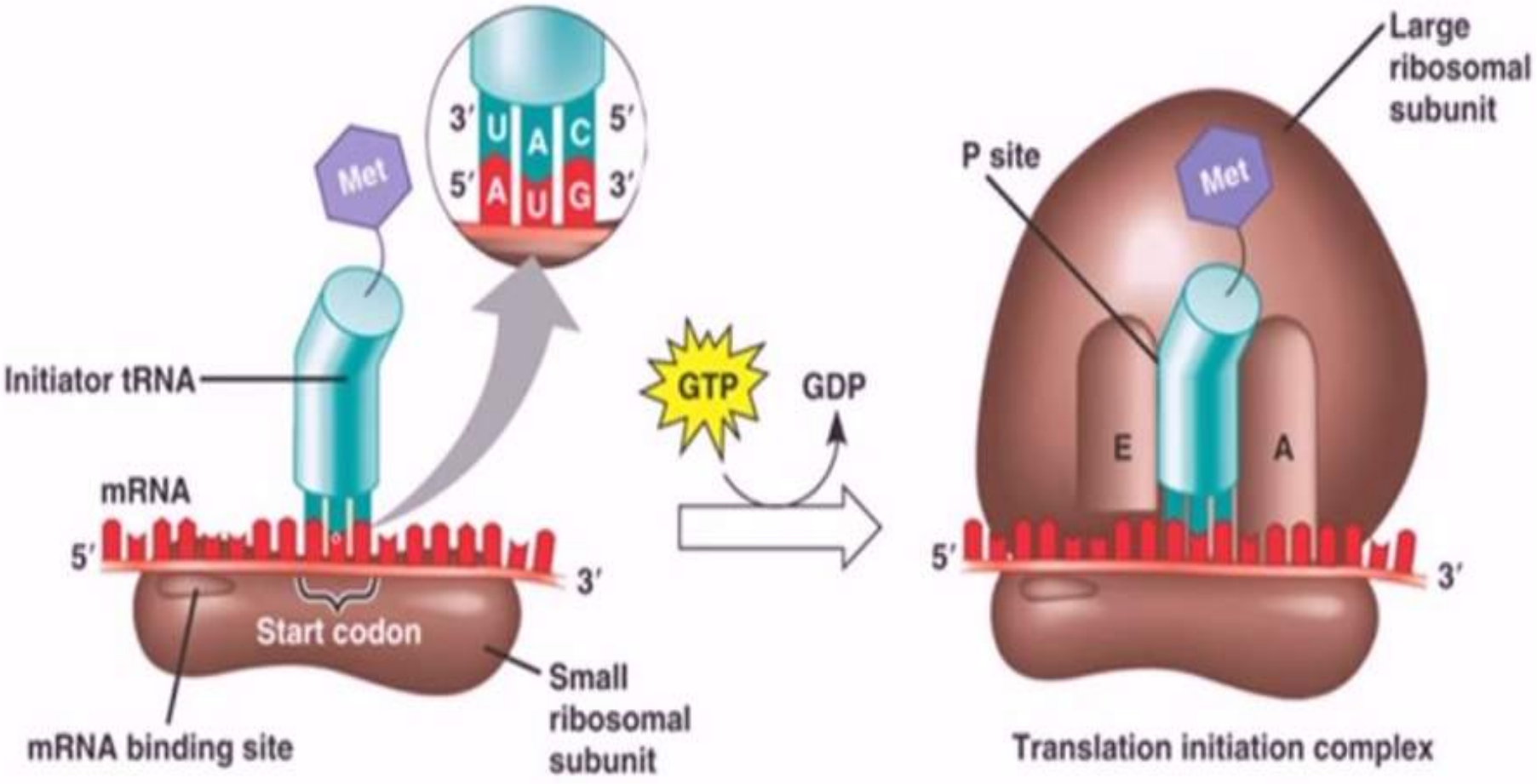



Figure 24.15: fMet-tRNAf has unique features that distinguish it as the initiator tRNA.

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# Translation: Initiation



# First Codon and the First Aminoacyl-tRNA

- Prokaryotic initiation codon is:
  - Usually AUG
  - Can be GUG
  - Rarely UUG
- Initiating aminoacyl-tRNA is *N*-formyl-methionyl-tRNA
- *N*-formyl-methionine (fMet) is the first amino acid incorporated into a polypeptide
- This amino acid is frequently removed from the protein during maturation

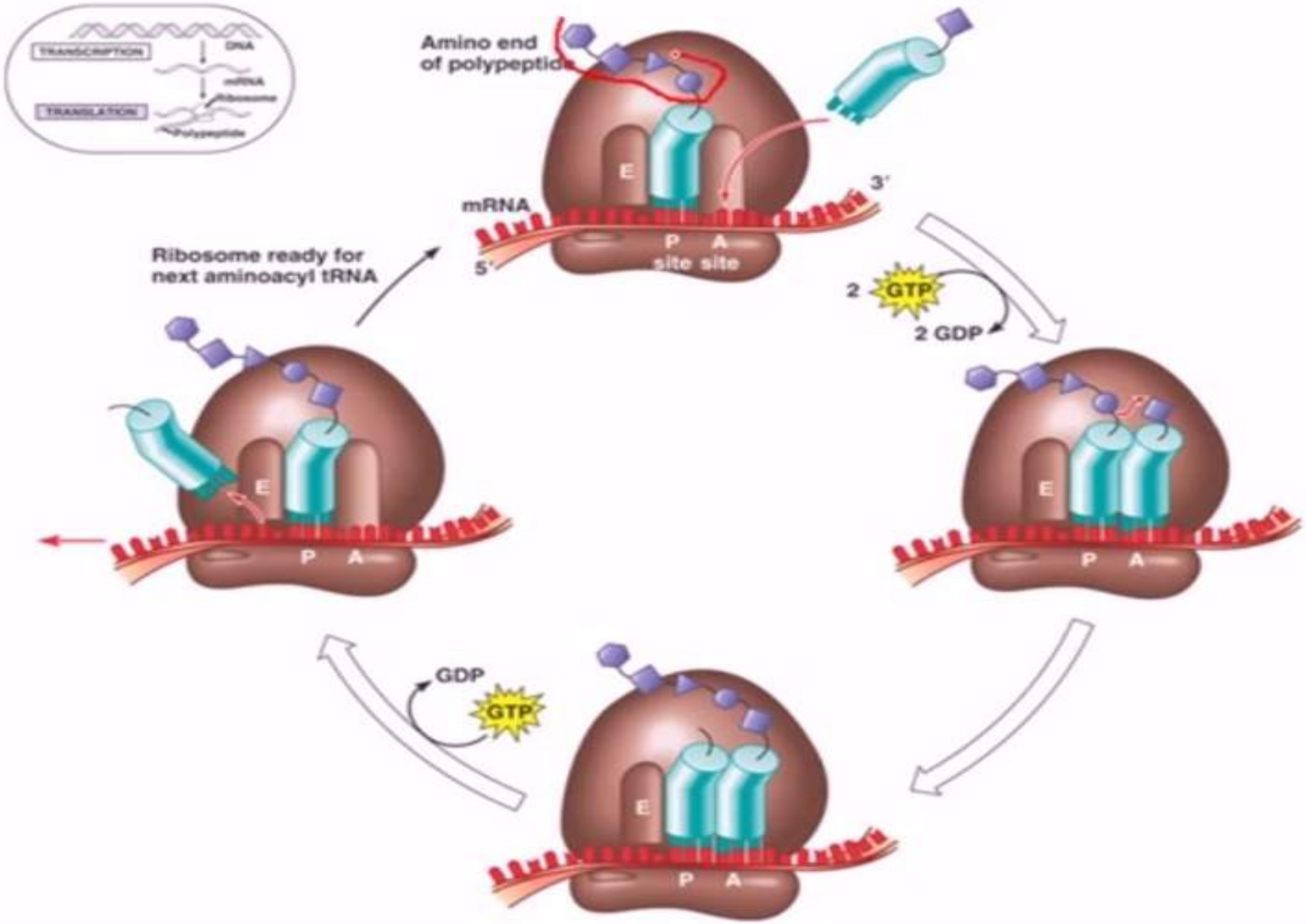
# INITIATION OF TRANSLATION

- It occurs in three steps

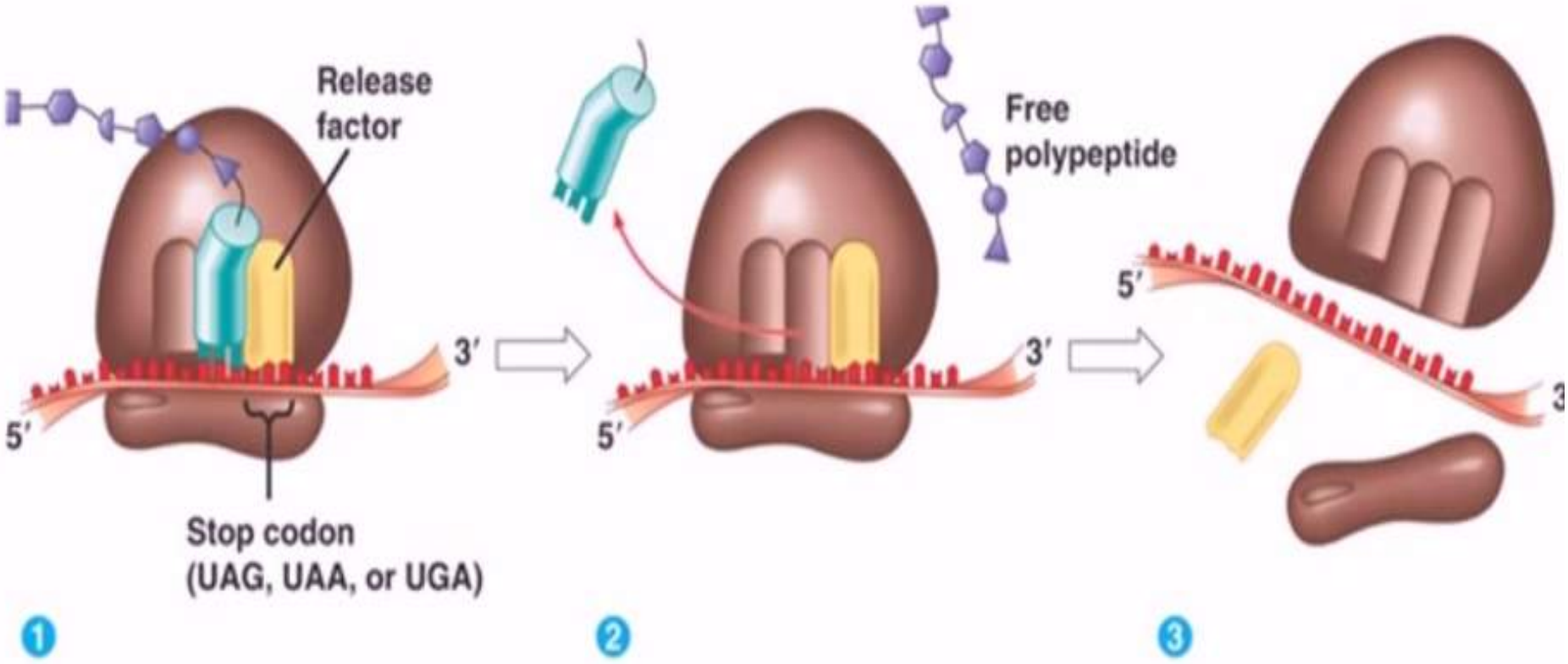
1) Ribosome must be recruited to the mRNA
2) Charged tRNA must be placed into the P site of ribosome
3) Ribosome must be precisely positioned over the start codon

- The initiator tRNA is charged with **N-Formyl methionine** in prokaryotes and with **methionine** in eukaryotes.
- Three initiation factors direct the assembly of an initation complex that contains mRNA and the initiator tRNA called translational initiation factors.

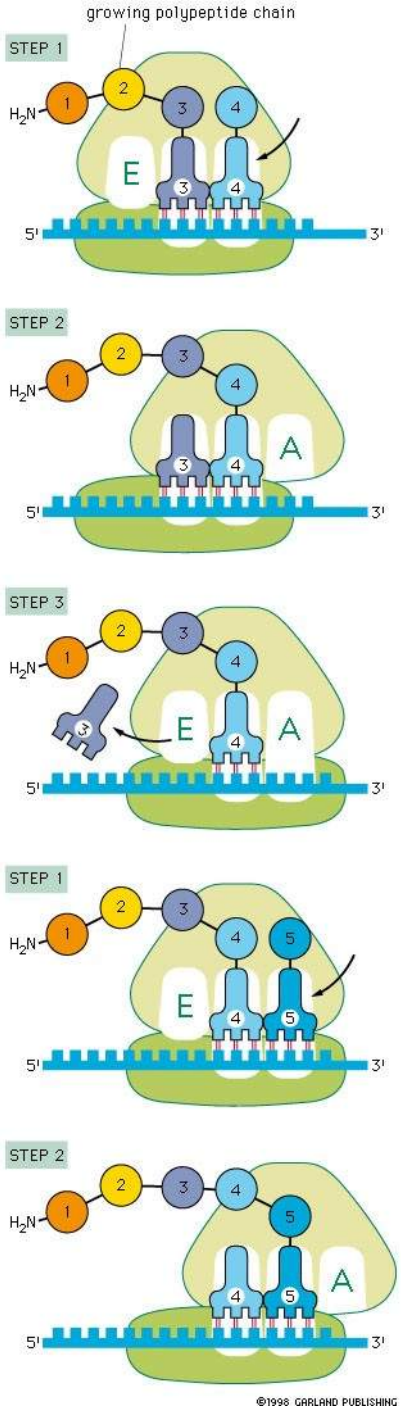
# Translation: Elongation



# Translation: Termination



# mRNA translation mechanism



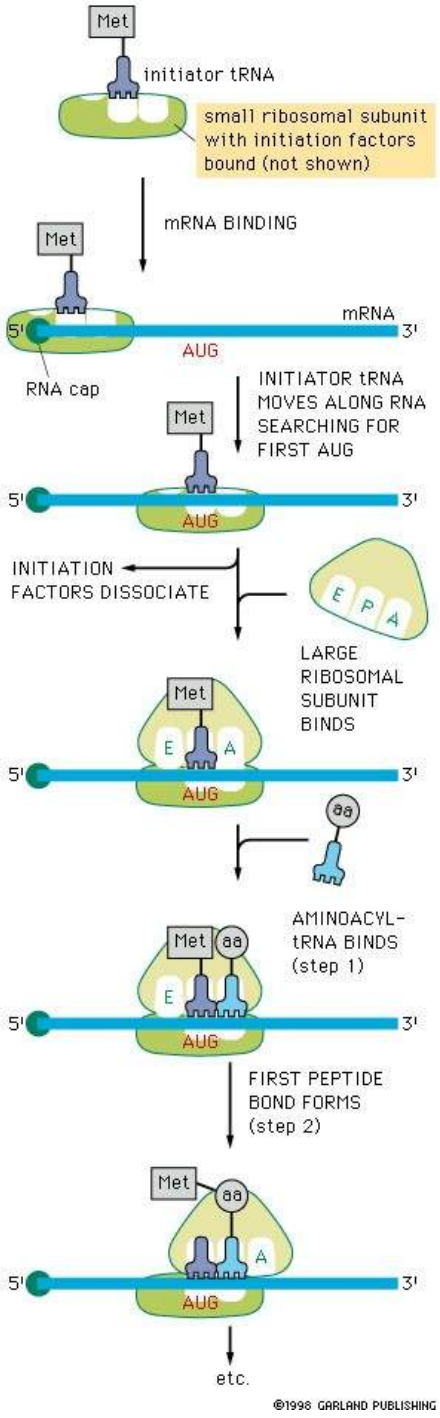

**Step1:** An aminoacyl-tRNA molecule binds to the A-site on the ribosome

**Step2:** A new peptide bond is formed

**Step3:** The small subunit moves a distance of three nucleotides along the mRNA chain ejecting the spent tRNA molecule

**Step4:** The next aminoacyl-tRNA molecule binds to the A-site on the ribosome

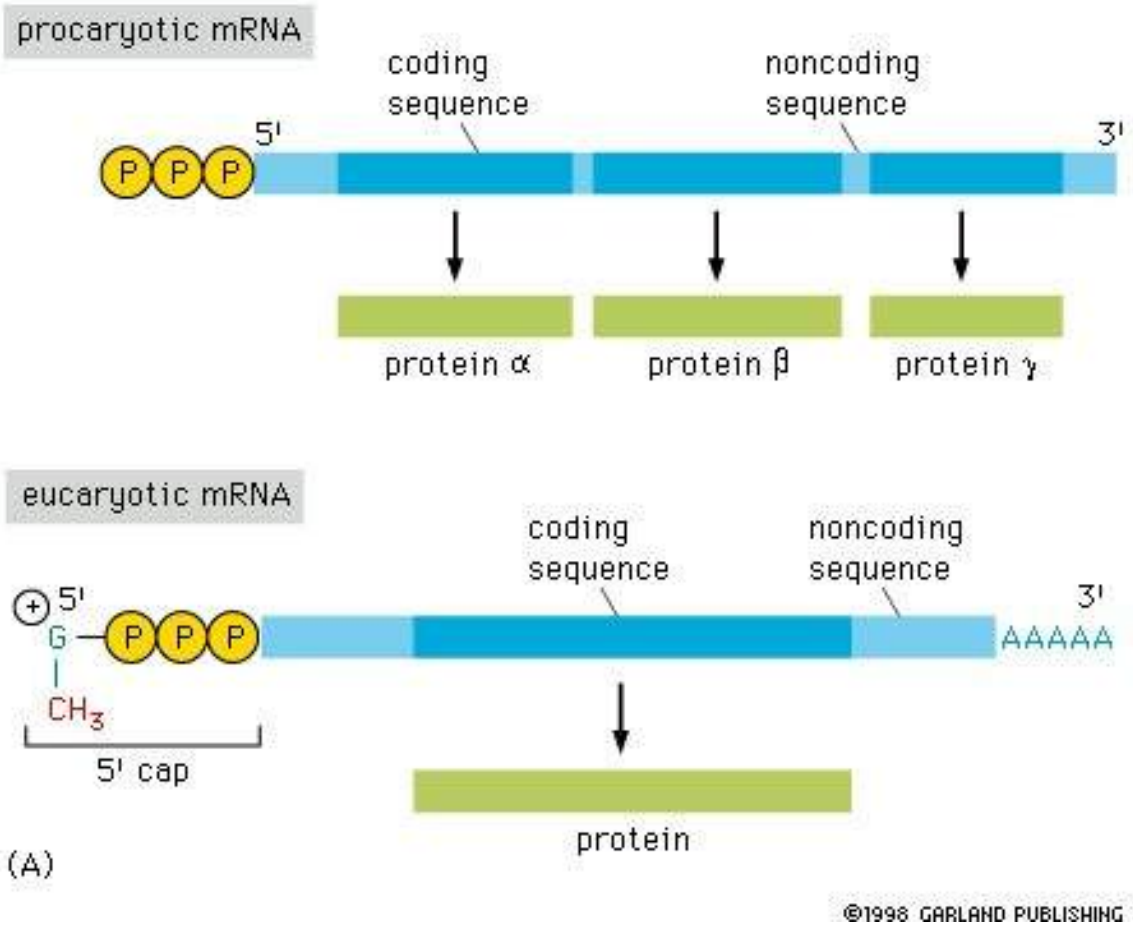
**Step5:** . . .



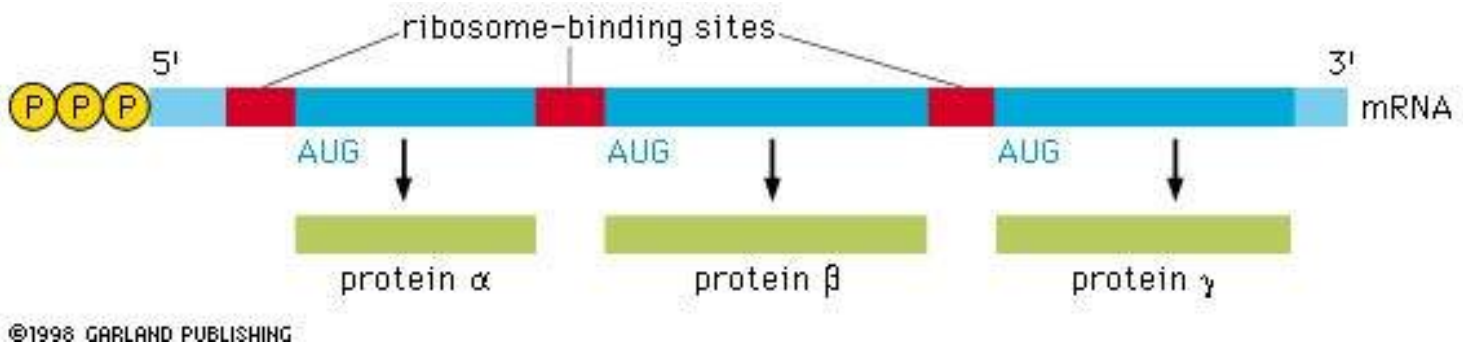

# The initiation phase of protein synthesis in eukaryotes

1. **Initiation complex (small ribosomal subunit + initiation factors) binds DNA and searches for start codon**
2. **Large ribosomal subunit adds to the complex**
3. **Translation starts**
4. **. . .**

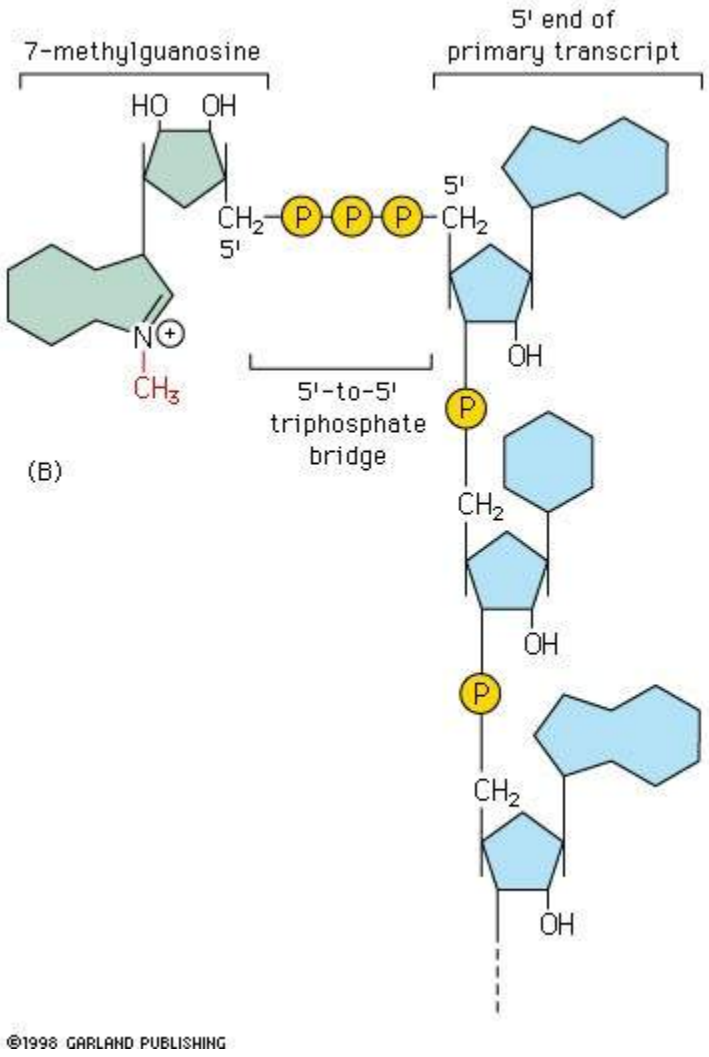
# Prokaryotic vs eukaryotic mRNA molecules

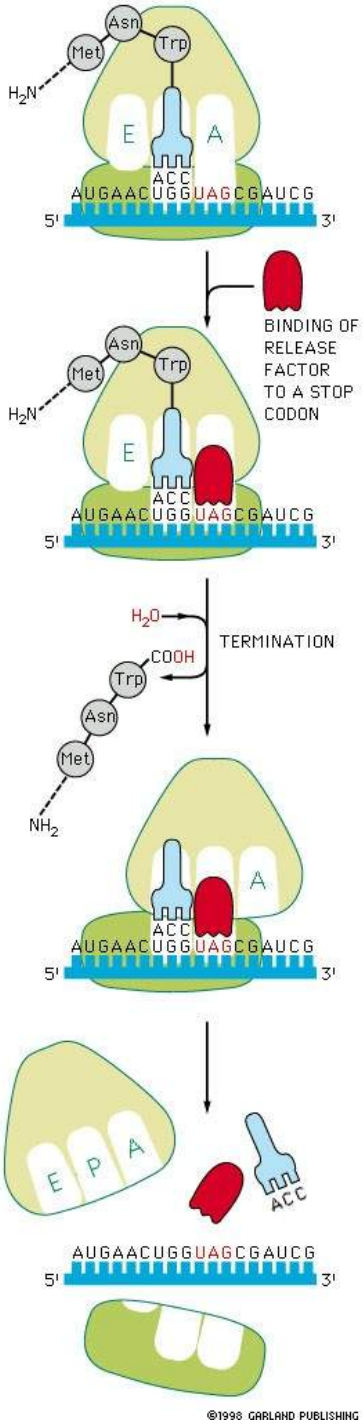


# Structure of a typical prokaryotic mRNA molecule



# 5´ end capping of eukaryotic mRNA molecules






# The final phase of protein synthesis

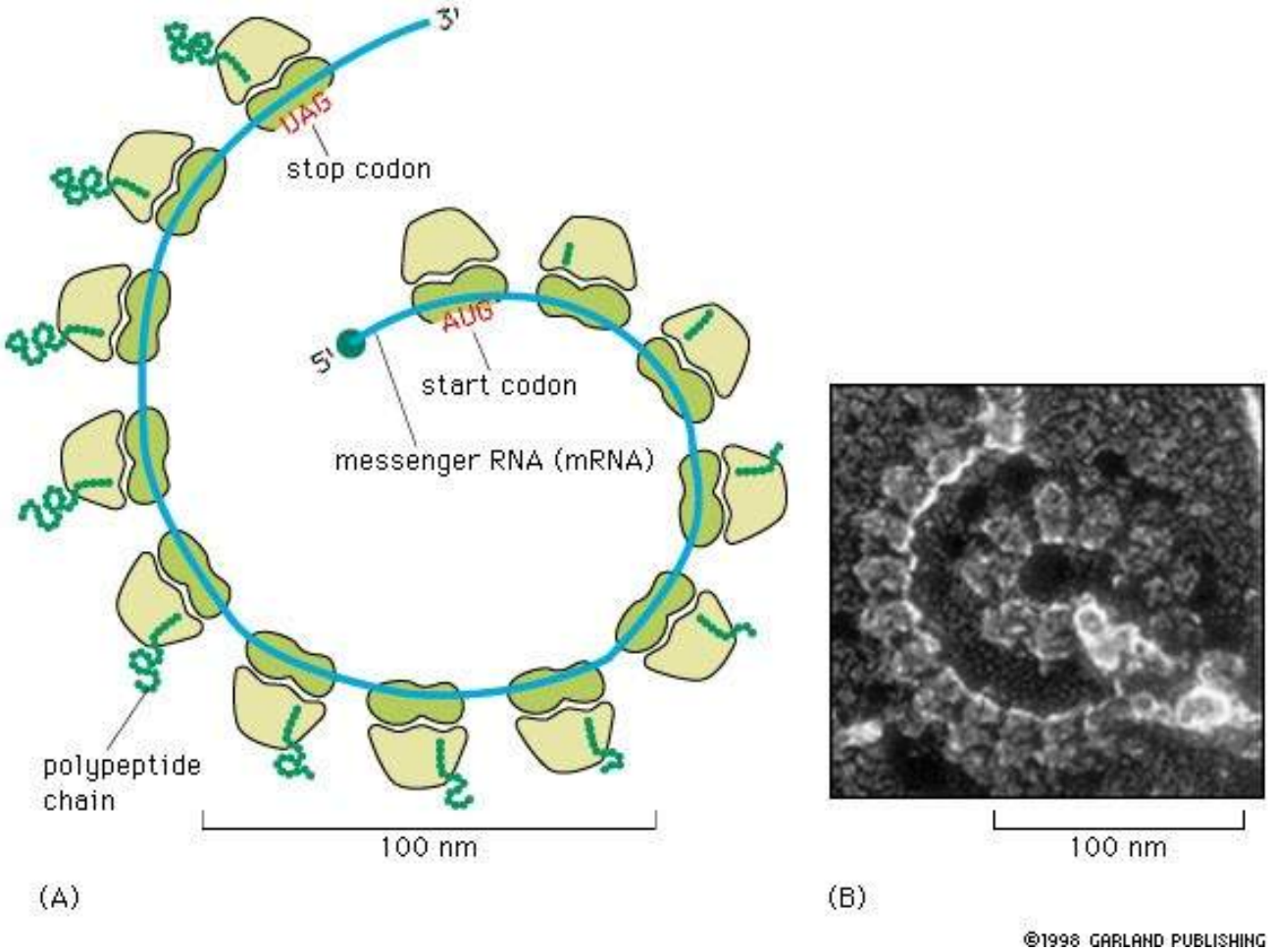
→ binding of **release factor** to a **stop codon** terminates translation

→ the completed polypeptide is released

→ the ribosome dissociates into its two separate subunits

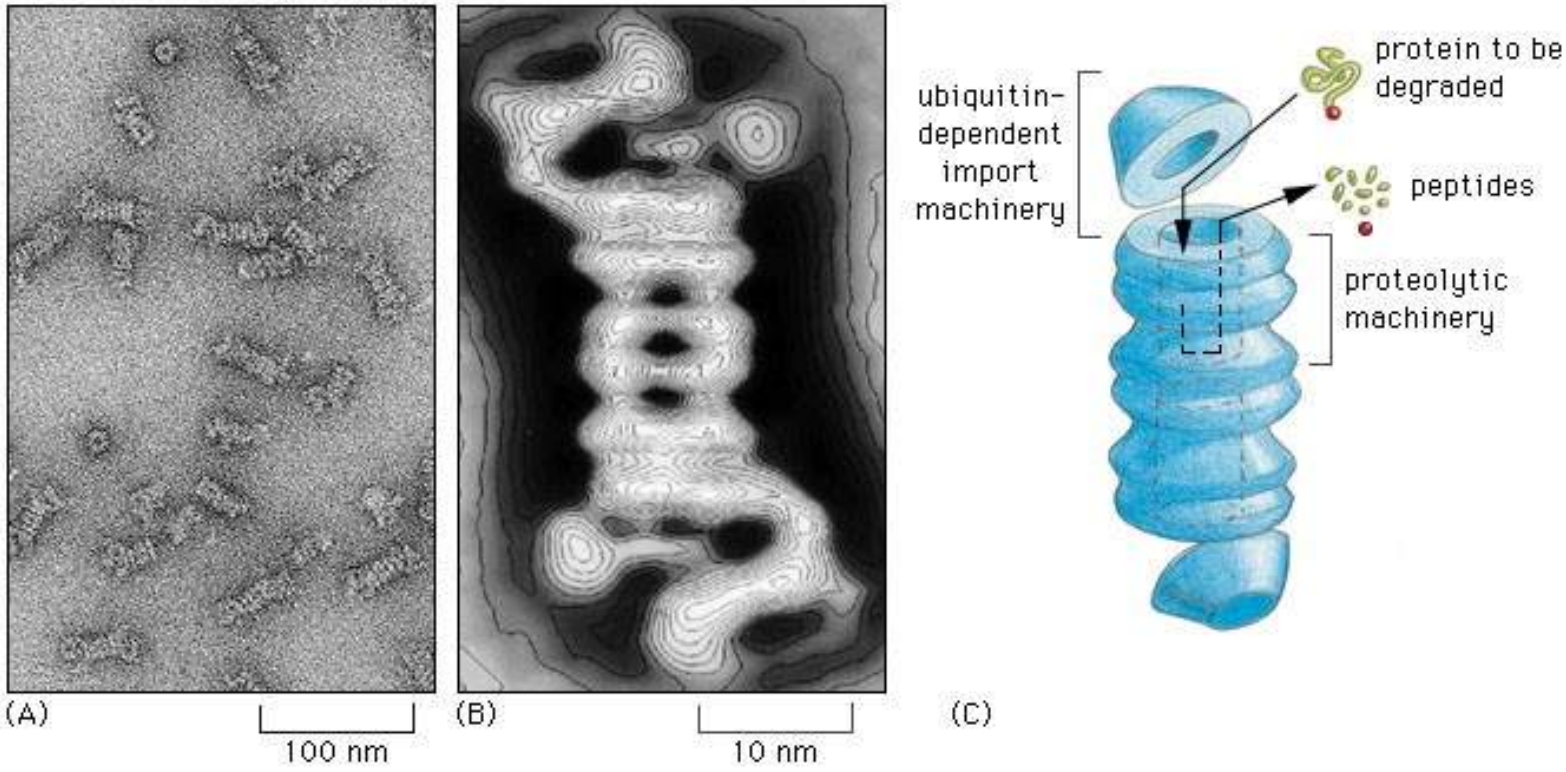
# Polyribosomes:

## several ribosomes can simultaneously translate the same mRNA molecule

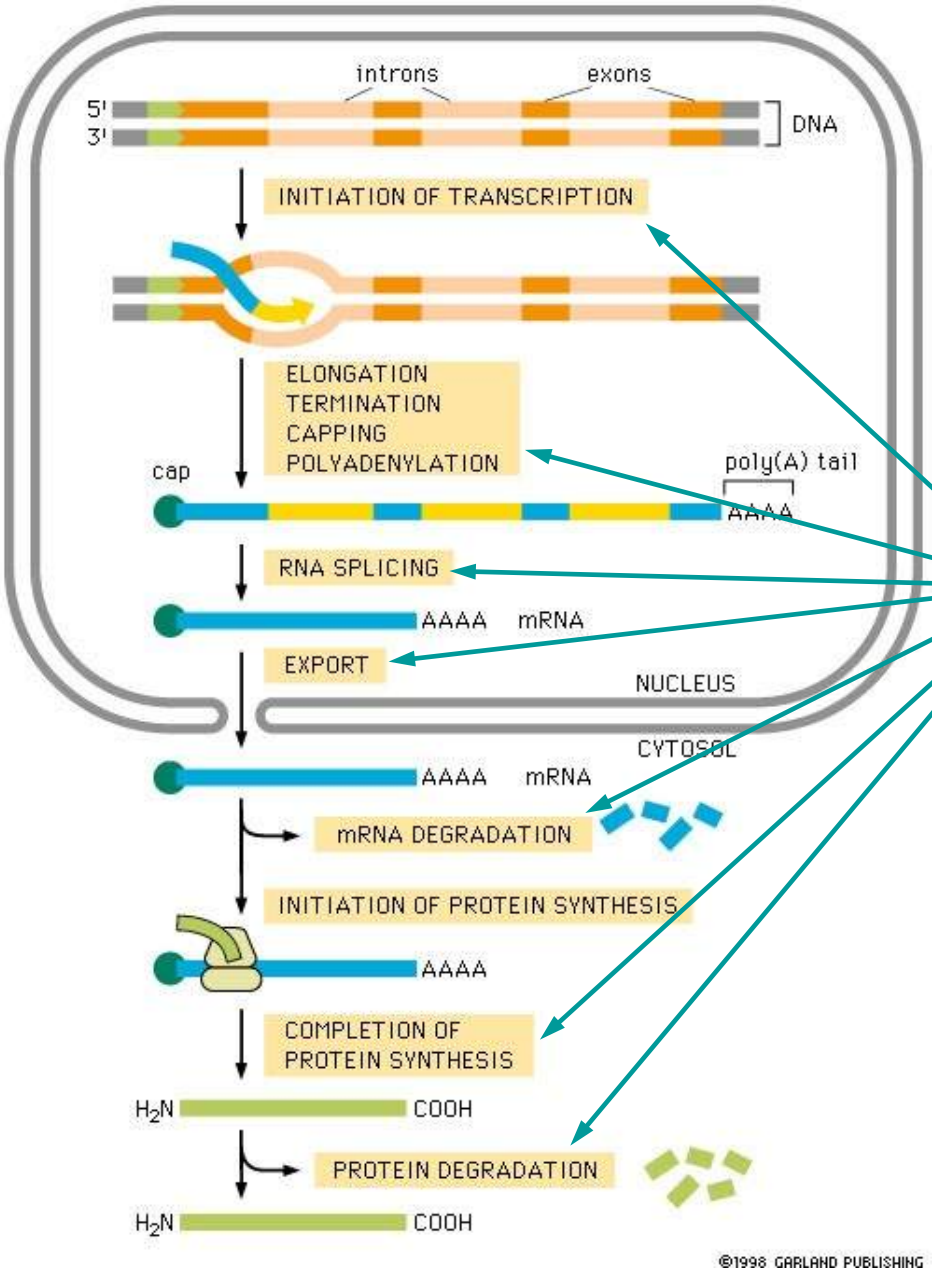


# Proteasomes:

## degradation of “unwanted” proteins in eukaryotic cells

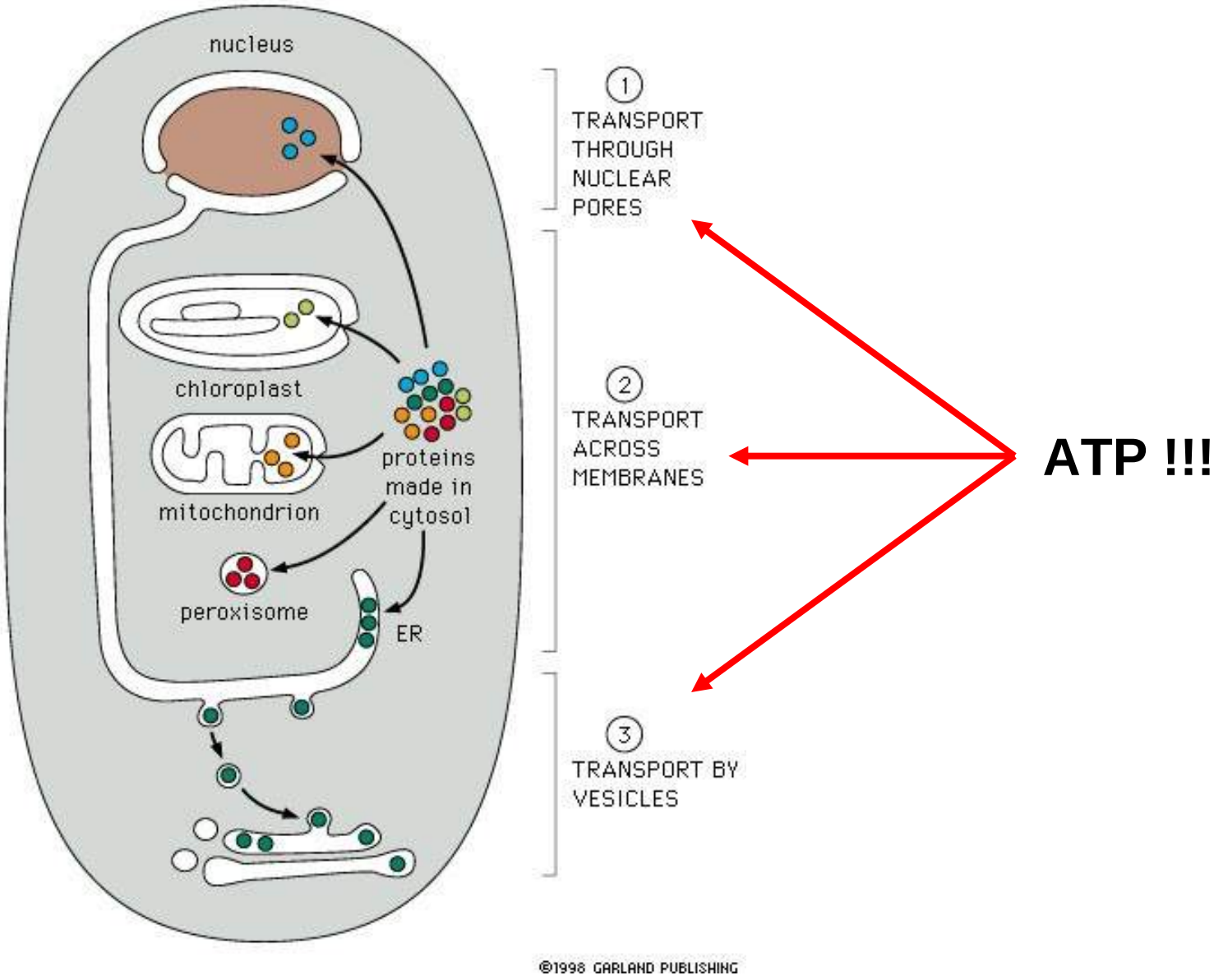


# The production of a protein by a eukaryotic cell



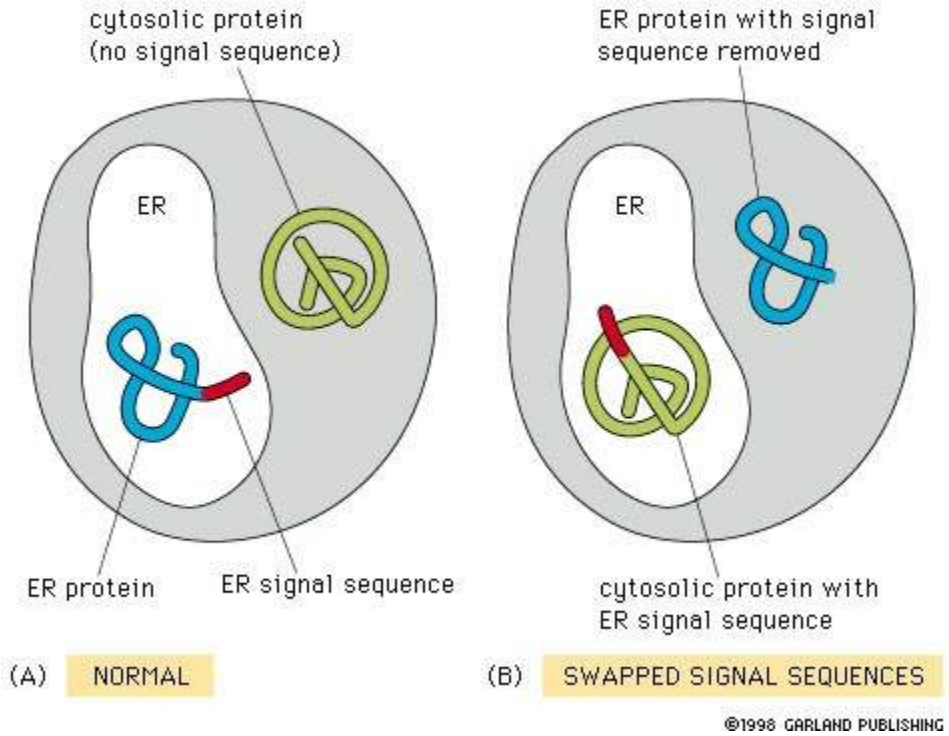

**Many levels of regulation/variation**

# Protein import by membrane-bounded organelles

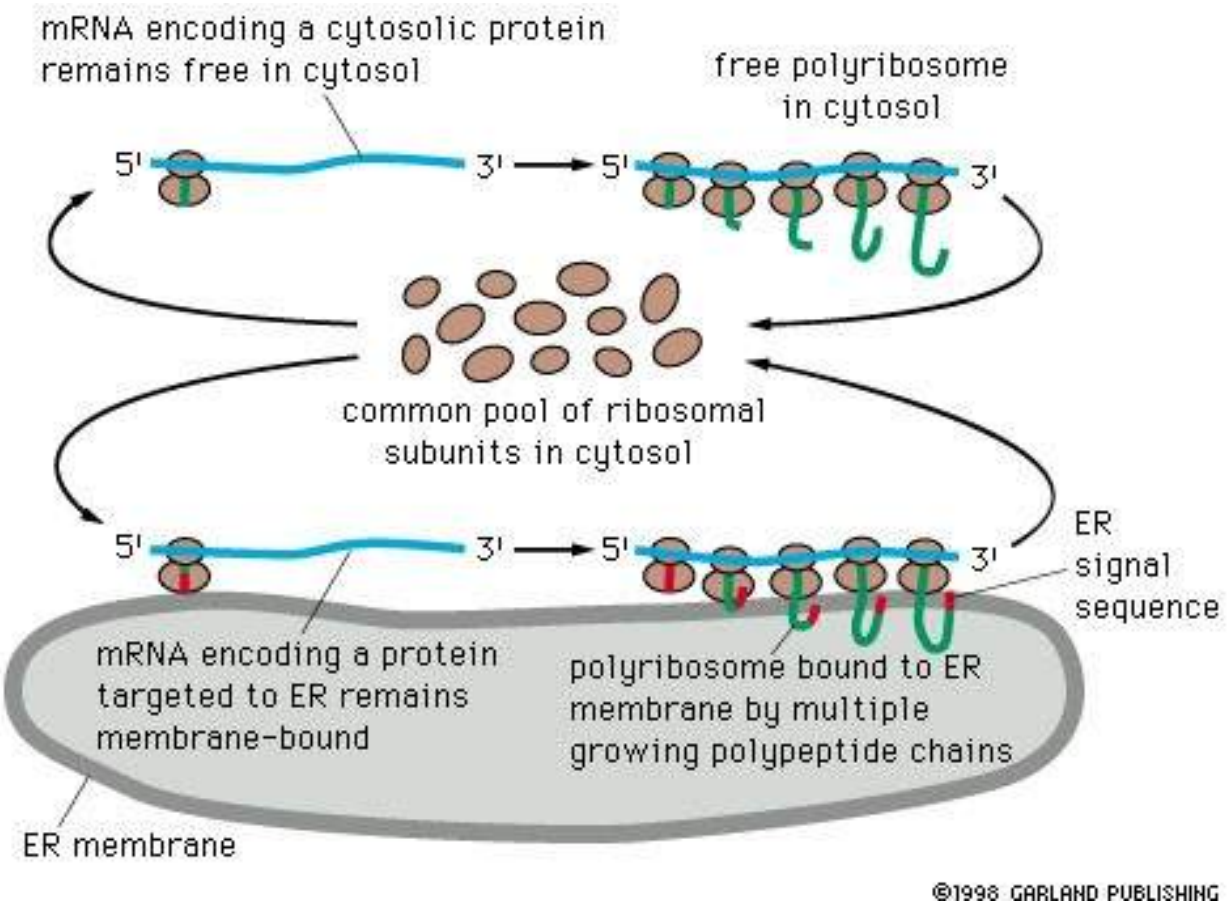


# The role of signal sequences in protein sorting

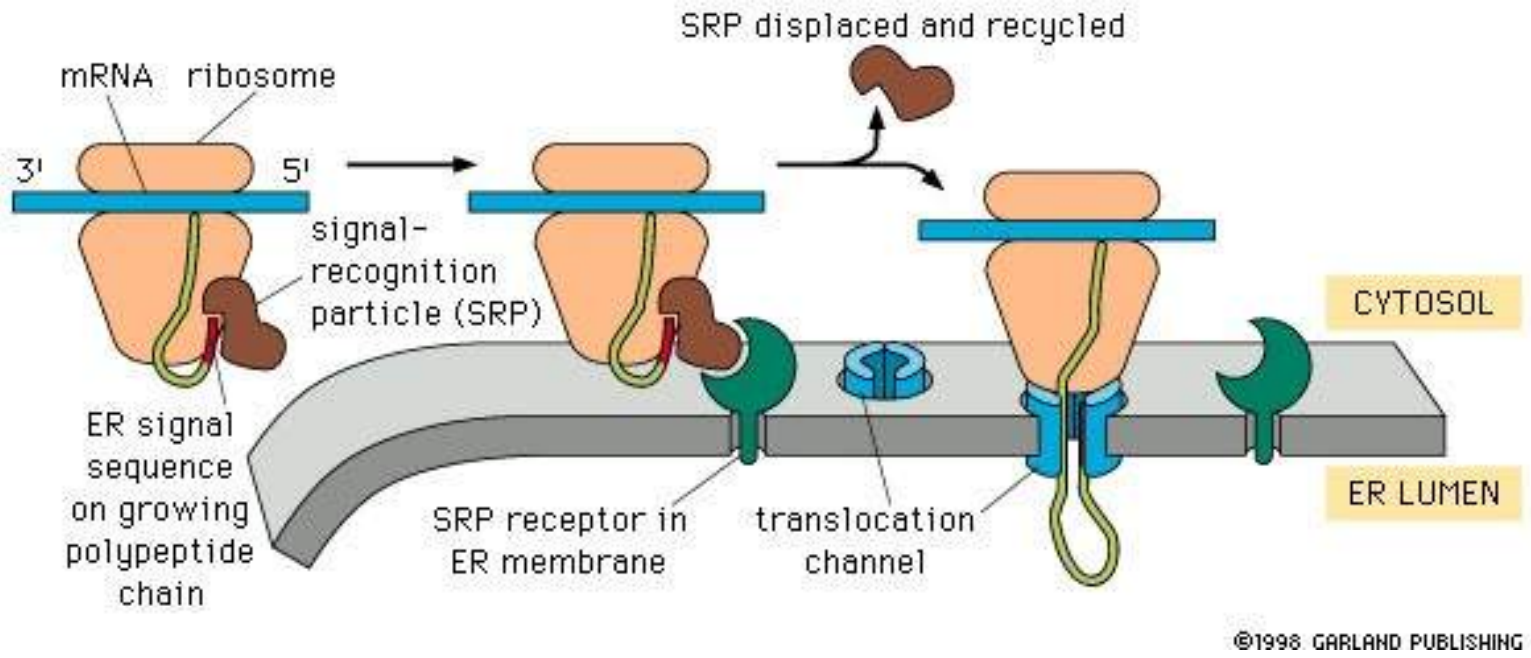
- ✓ **proteins destined for the ER: N-terminal signal sequence that directs them**
- ✓ **proteins destined to remain in the cytosol: no signal sequence**



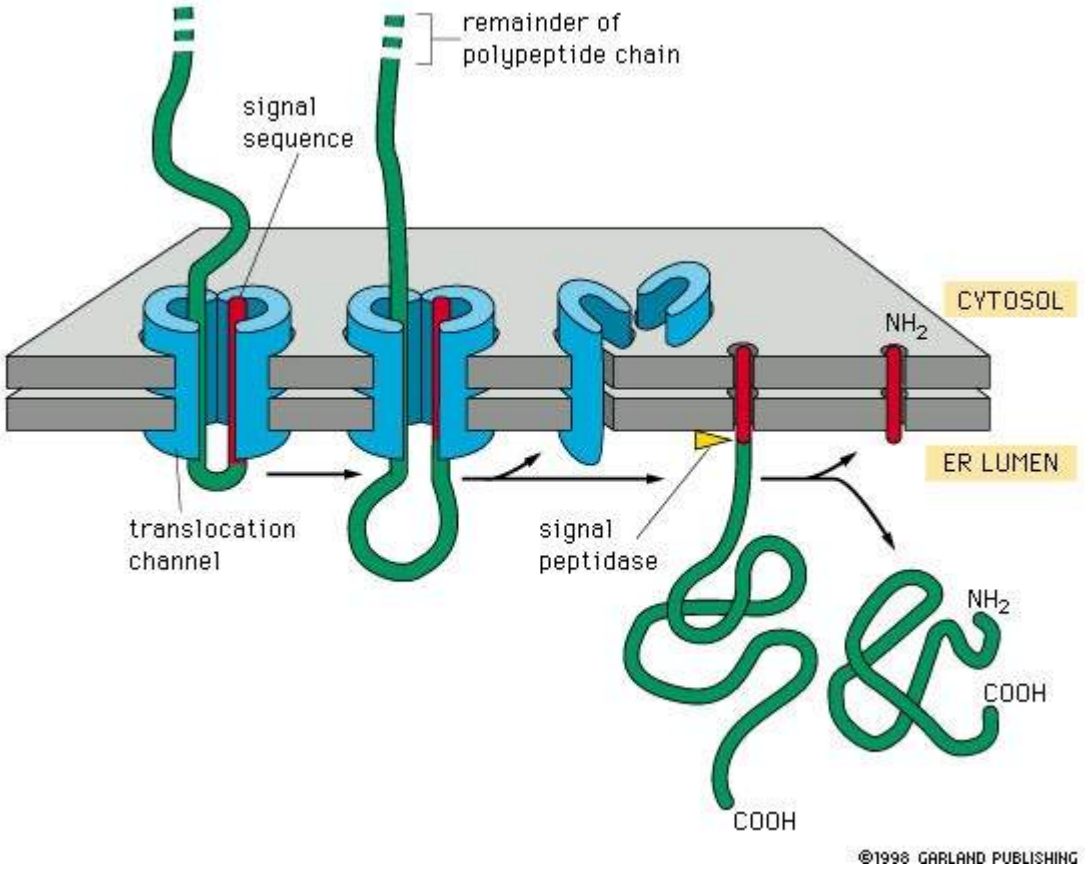
# Free and membrane-bound ribosomes



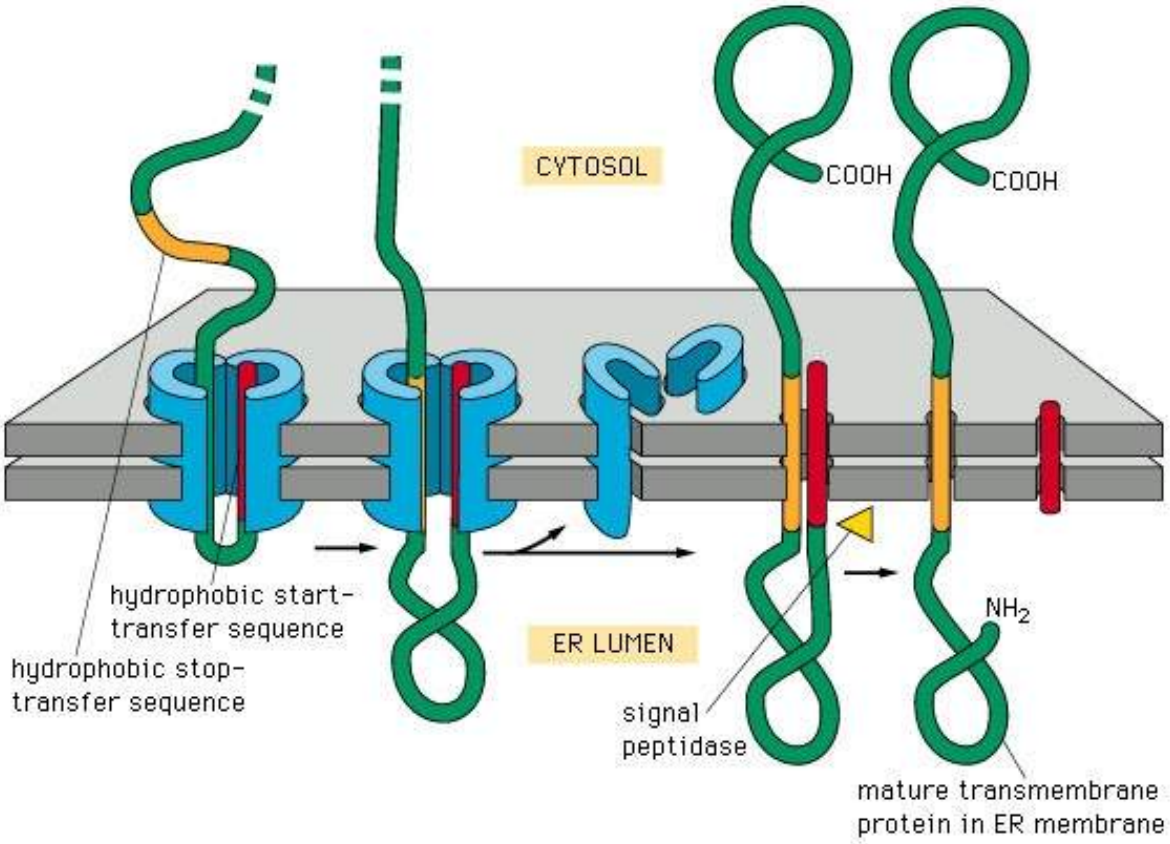
# ER signal sequence and SRP:
## directing ribosomes to the ER membrane



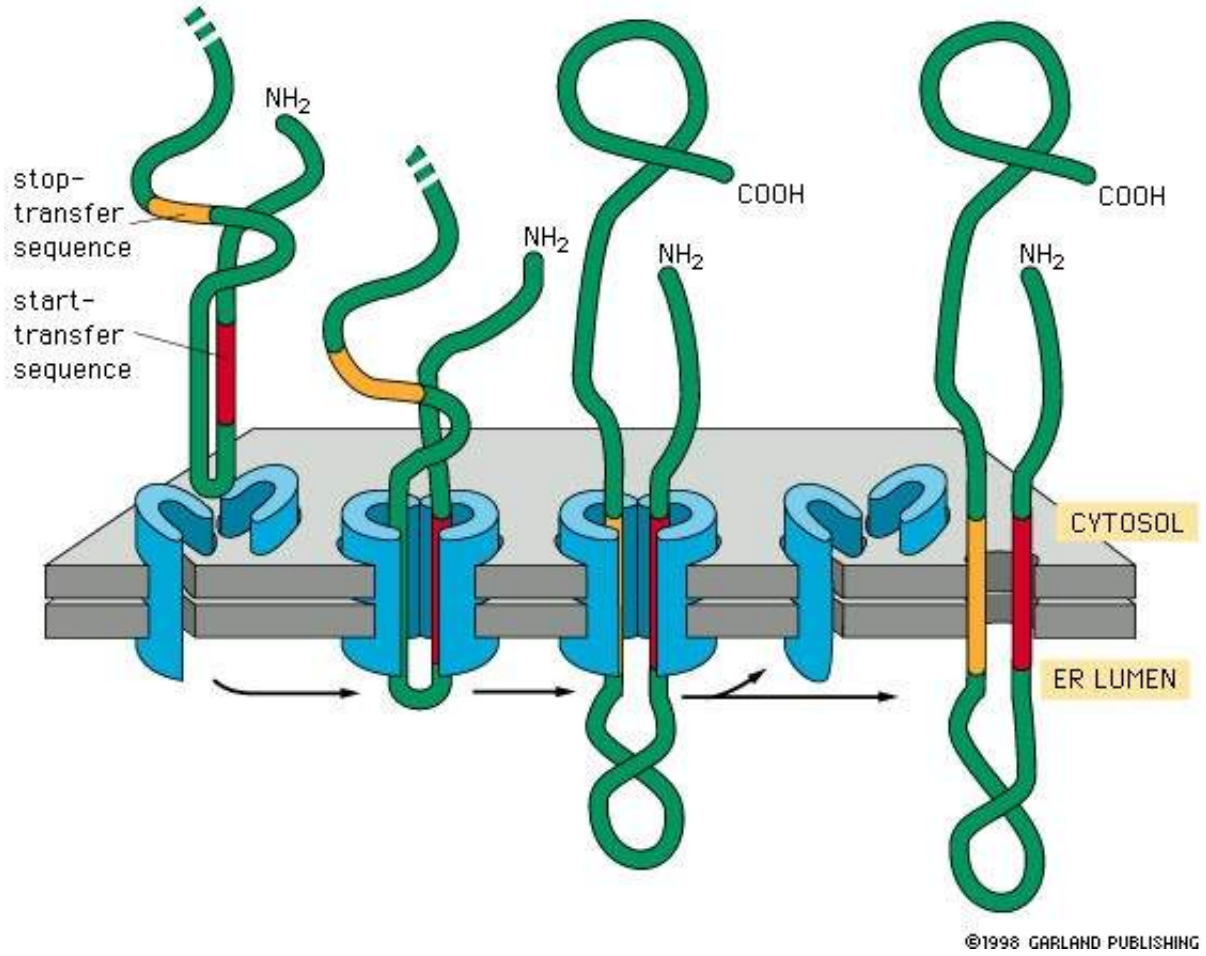
# Translocation of a soluble protein across the ER membrane



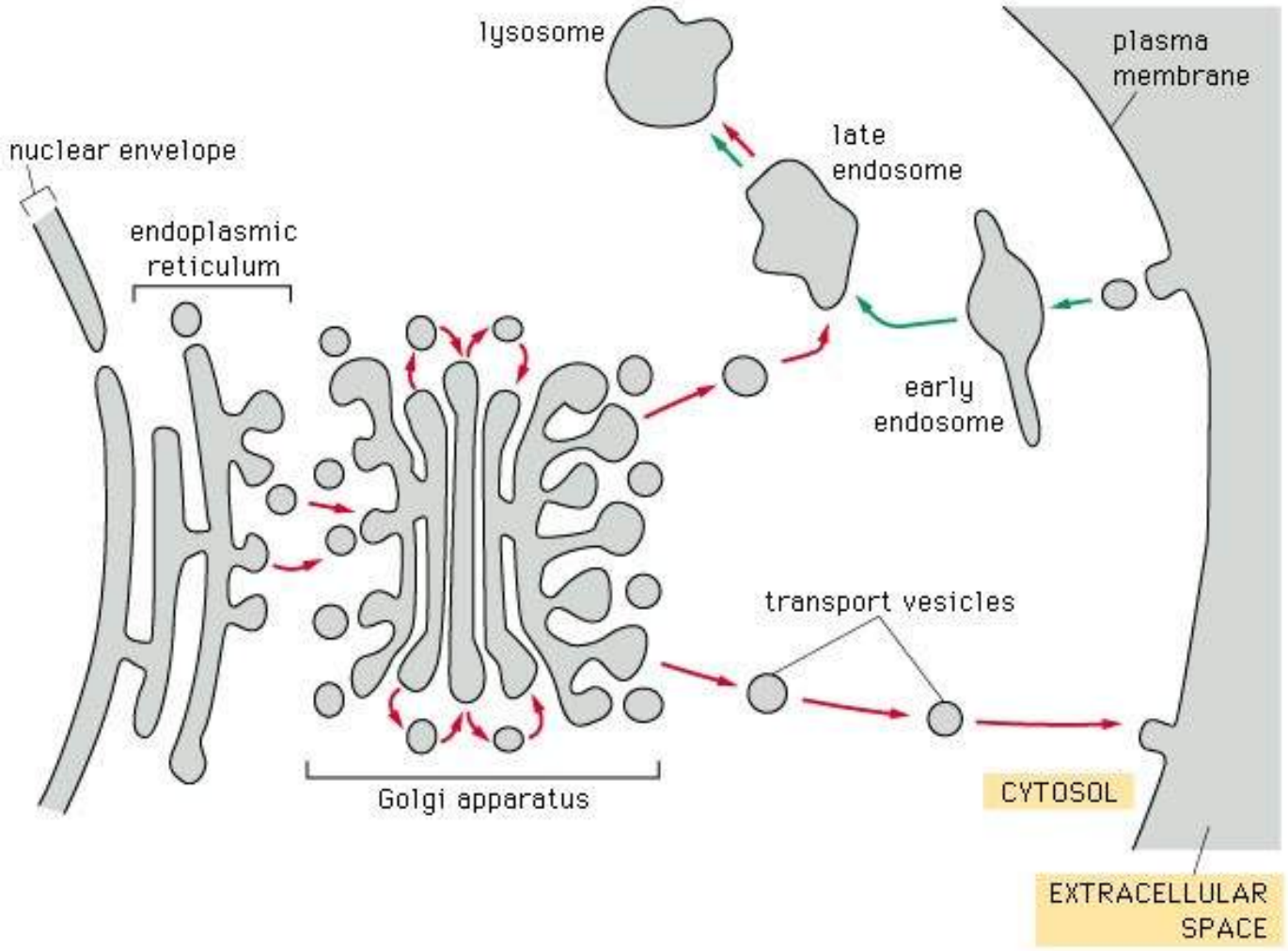
# Integration of a transmembrane protein into the ER membrane



# Integration of a double-pass transmembrane protein into the ER membrane

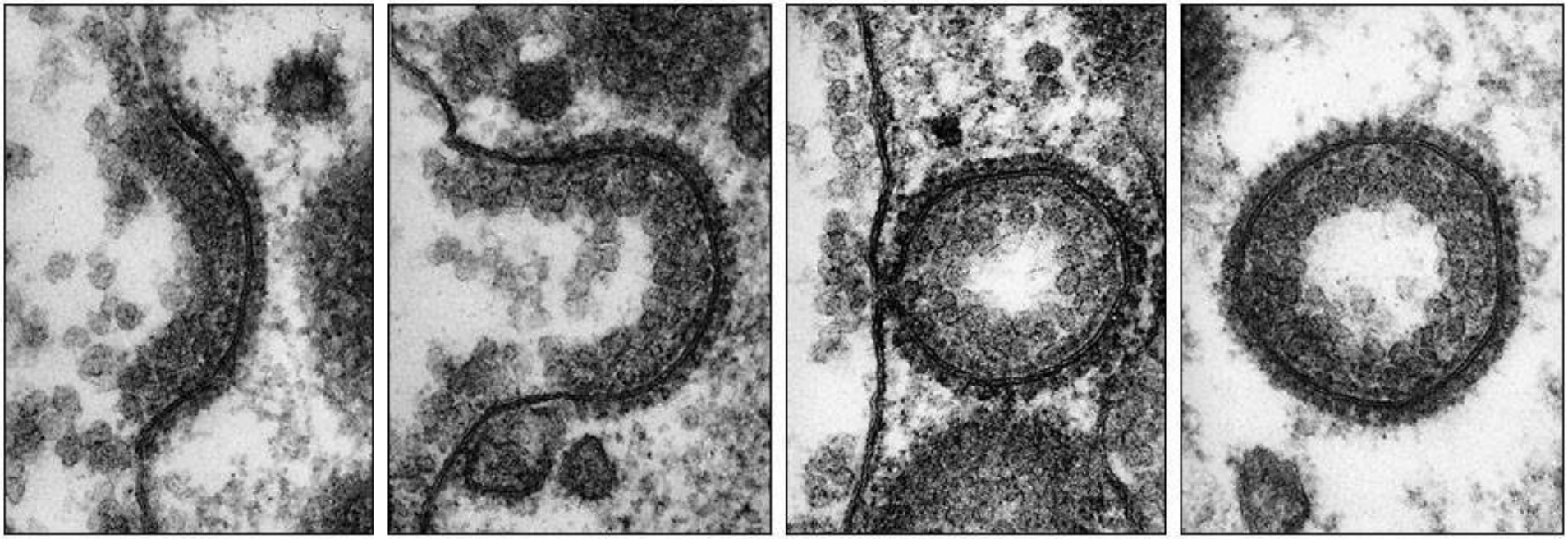


# Vesicular traffic

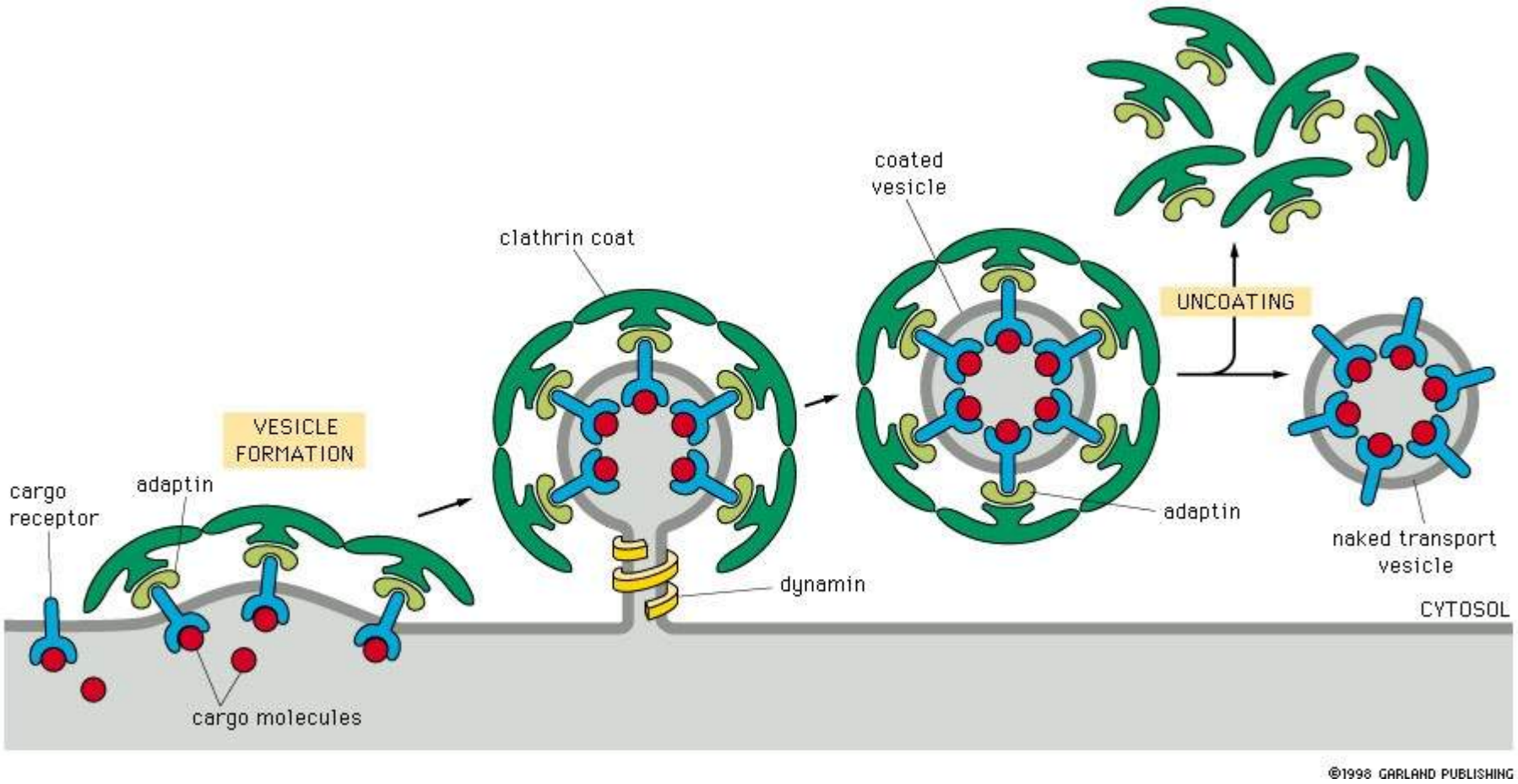



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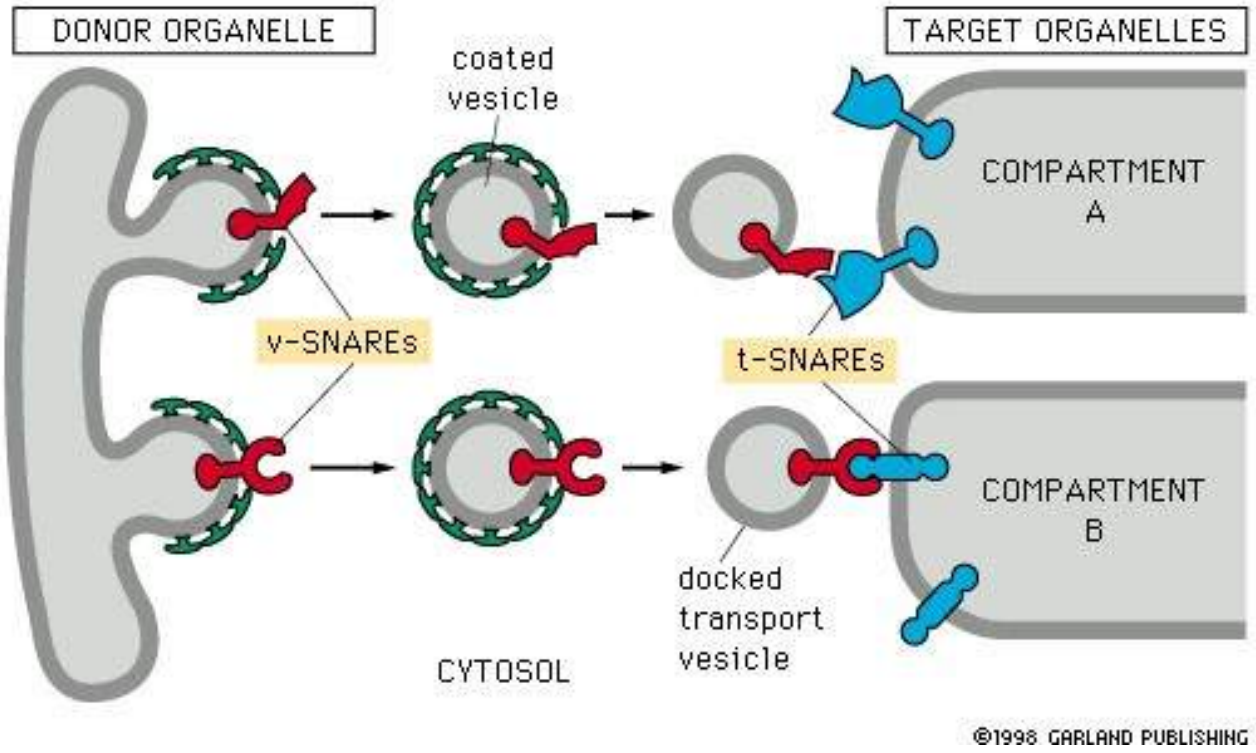
# Clathrin-coated pits and vesicles



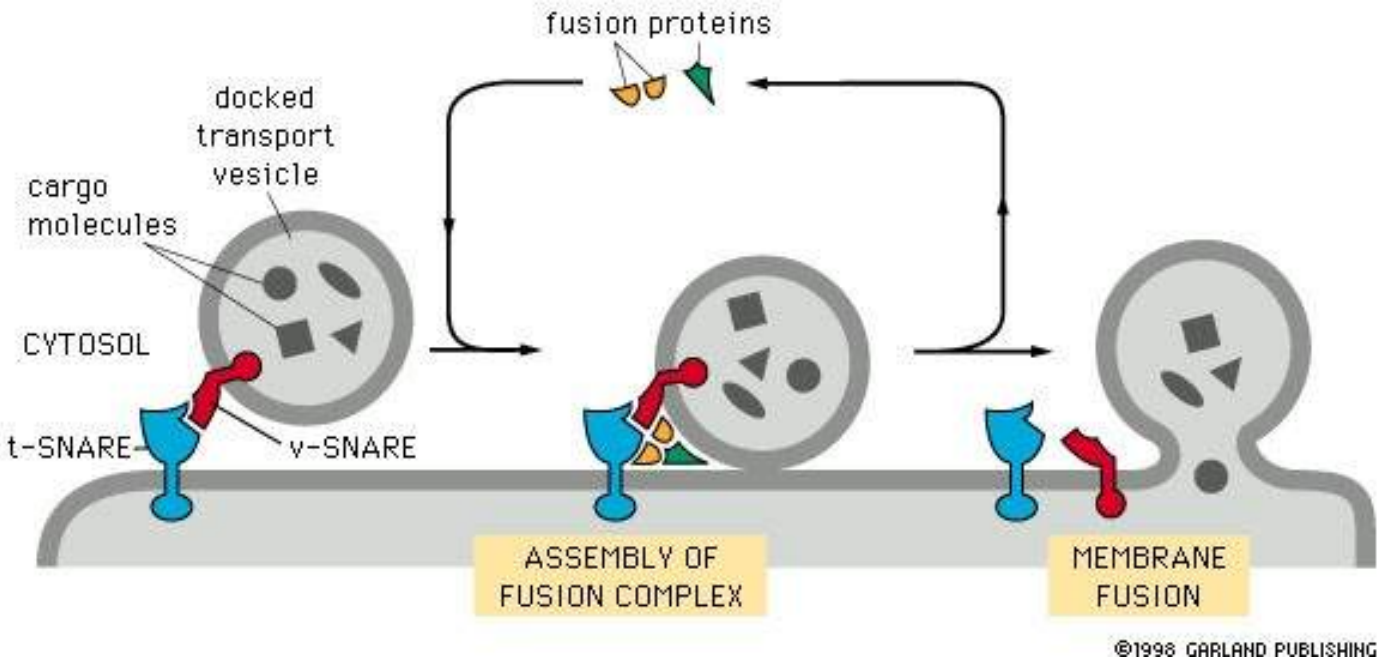
# Selective transport: mediated by clathrin-coated vesicles



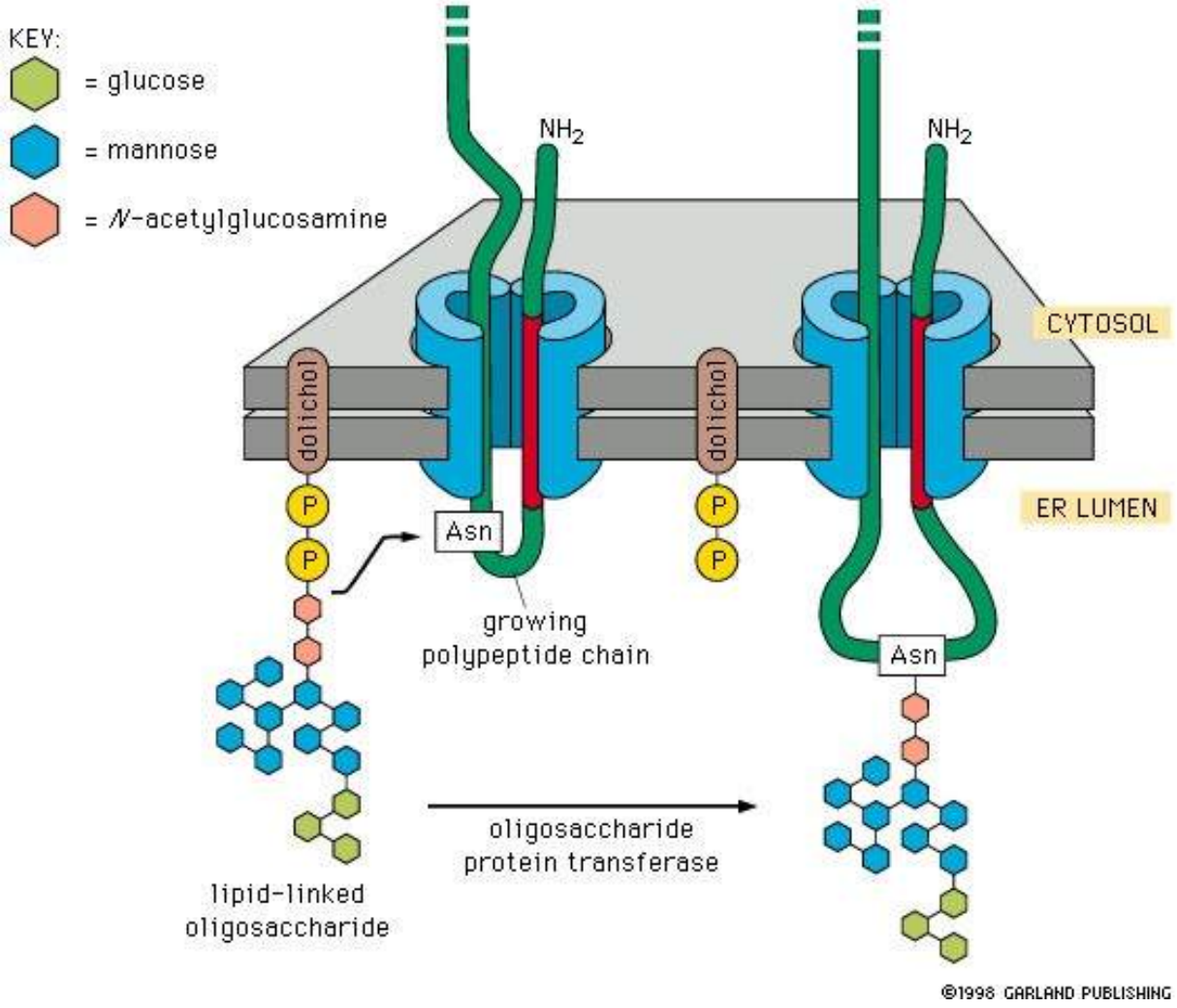
# Model of transport vesicle docking



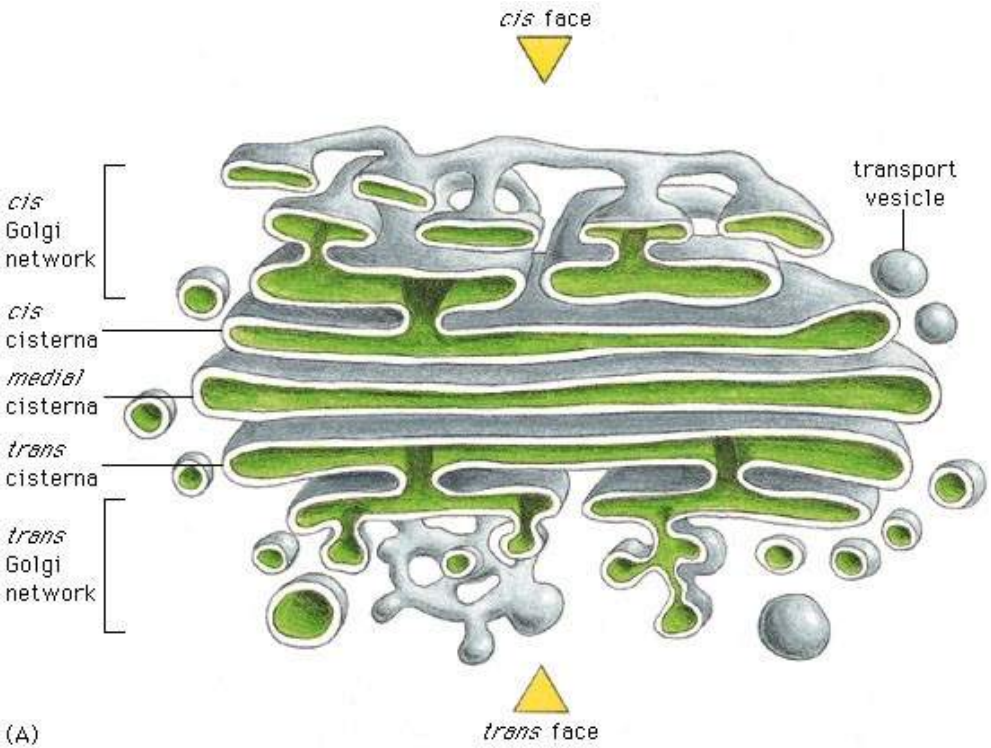
# Transport vesicle fusion

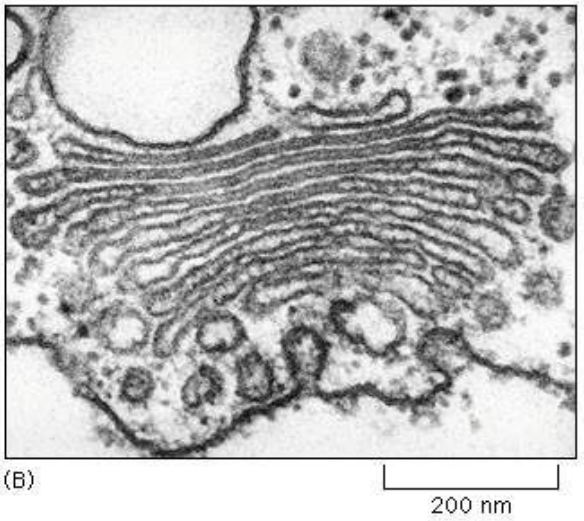


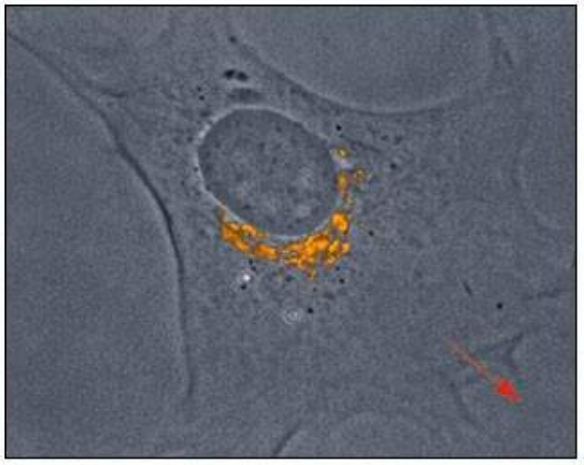
# Protein glycosylation in the ER



# The Golgi apparatus

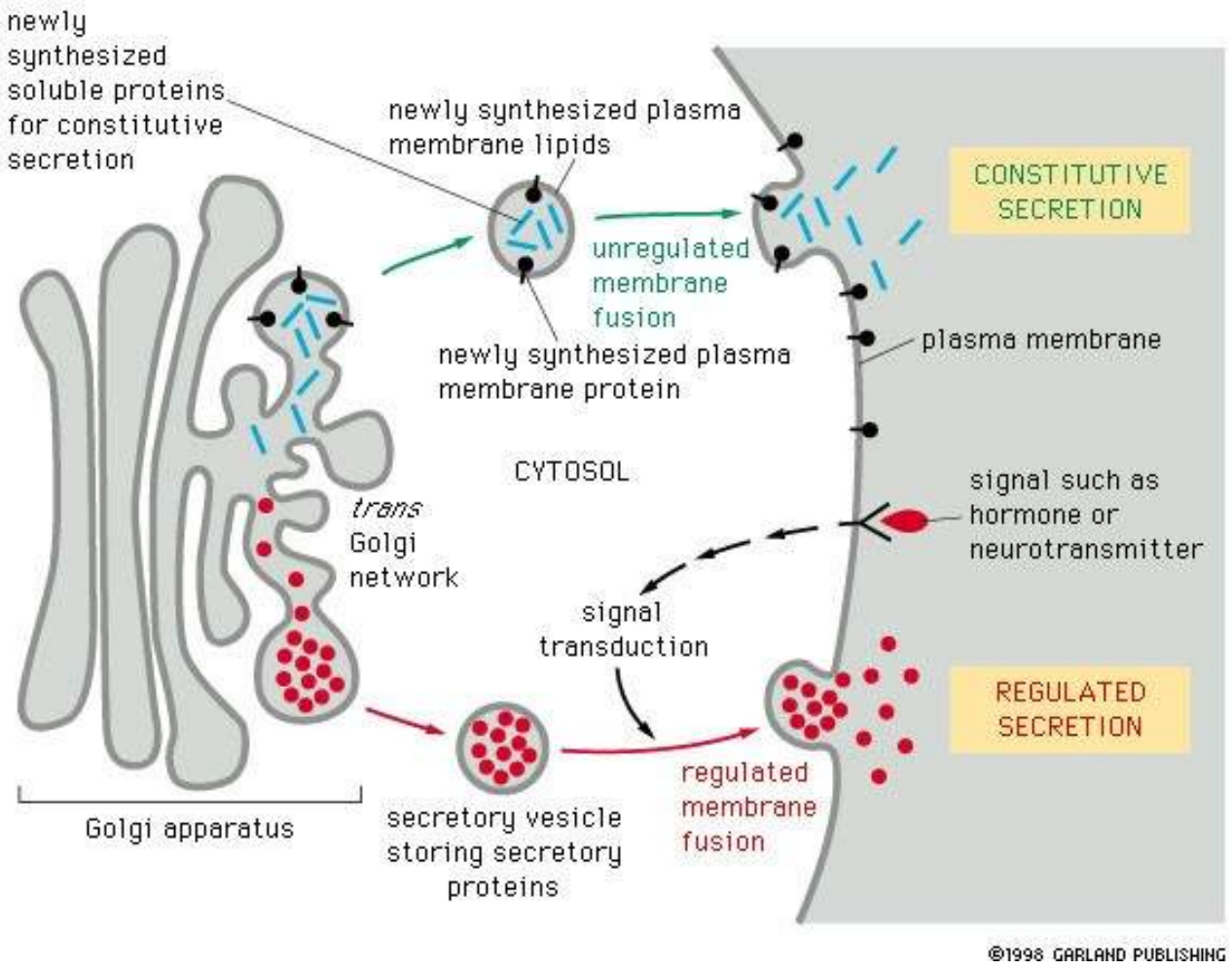


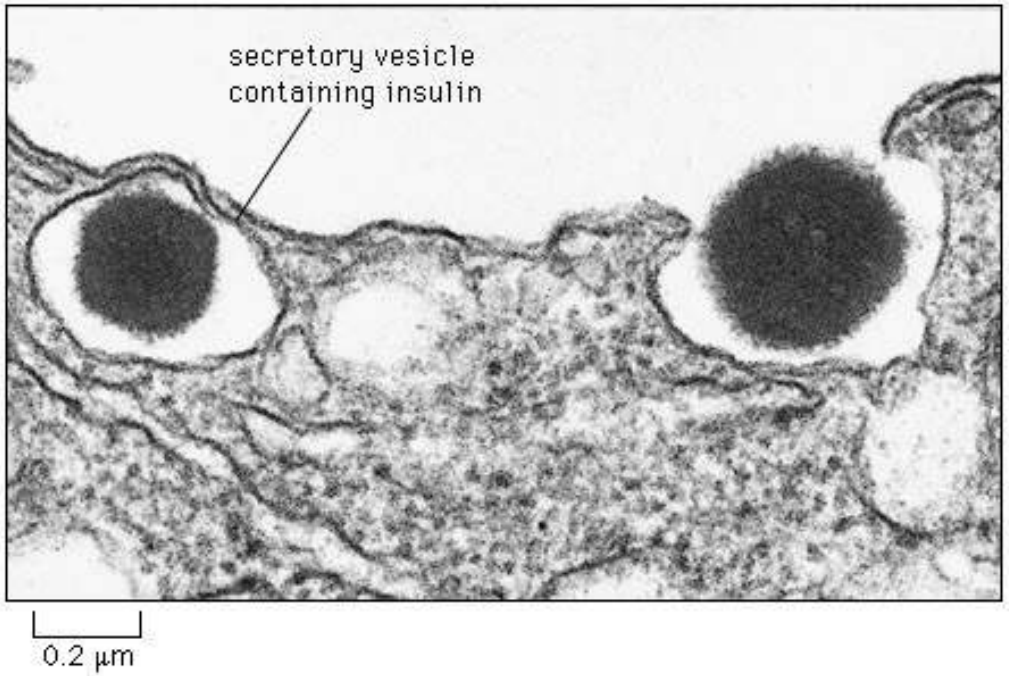




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# Exocytosis:
# the regulated and constitutive pathways



# Exocytosis of secretory vesicles




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# Antisense DNA Prevents Translation

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